

The Algorithmic Liquidity Trap: High-Velocity Swiping vs. Deep Engagement in Short-Video Platforms^{*}

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Abstract

Short-video platforms balance user retention with high-frequency impression generation. We model this ecosystem using a Mean-Field Stackelberg Game with “traffic recycling,” where users constantly re-enter the content pool. Unlike much of the theoretical literature that relies solely on stylized parameters, we calibrate the model’s micro-foundations using KuaiRand—a large-scale randomized experiment dataset from Kuaishou—ensuring that user behavioral parameters reflect real-world attention dynamics. Solving the coupled HJB and Non-Local Fokker-Planck equations via a novel Spectral Soft Actor-Critic algorithm, we identify a critical “Algorithmic Liquidity Trap.” While unconstrained optimization mathematically maximizes swipe flux, it drives dwell times to a physiological floor well below the IAB/MRC viewability standard of 2 seconds. We term this “Phantom Liquidity”—high throughput with zero effective economic value. Furthermore, we reveal a “Dopamine Treadmill” effect: improved content quality paradoxically increases user impatience, explaining the difficulty of sustaining long-form content. Our results demonstrate that without explicit constraints—such as weights for completion rates and creator interactions—algorithmic maximization inevitably collapses the ecosystem, validating the necessity of current industry governance structures.

Keywords: short-video platforms, mean-field games, traffic recycling, deep reinforcement learning, algorithmic liquidity trap, platform governance.

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1 Introduction

Short-video platforms such as TikTok, Instagram Reels, and Kuaishou operate at a scale that makes their algorithmic design choices macroeconomic in consequence. Yet a striking paradox lies at the heart of their business model: the very optimization that maximizes a platform’s headline traffic metric—swipe frequency—may systematically destroy the economic value of that traffic. Under the industry-standard Media Rating Council (MRC) viewability rule, a mobile video advertisement is billable only if it plays for at least two continuous seconds (Media Rating Council (MRC) and Interactive Advertising Bureau (IAB), 2015). An algorithm that drives dwell times below this physiological floor generates impressions that advertisers are not obligated to pay for. High nominal traffic thus coexists with near-zero effective ad revenue—a state we term *Phantom Liquidity*. The central question of this paper is: *Why does unconstrained algorithmic optimization inevitably produce this market failure, and what structural feature of the platform ecosystem makes it unavoidable?*

The answer, we argue, lies in a mechanism absent from existing models of user behavior and platform operations: *Traffic Recycling*. In classical models of consumer search (Ke, Shen, & Villas-Boas, 2016; Stokey, 2008), stopping is an absorbing state—a user who abandons a product exits the system. Short-video platforms violate this assumption structurally. When a user swipes away from a video, the platform immediately presents the next one; exits to competing platforms occur in fewer than 2% of swipe events in large-scale field data (Gao et al., 2022). This closed-loop topology means that the platform’s algorithm does not merely influence a single viewing decision but continuously reshapes the *macroscopic equilibrium* of the entire user population. The standard toolkit of optimal stopping theory, which characterizes each swipe as an independent exit decision, therefore misses the feedback mechanism through which individual swiping behavior aggregates into a systemic failure of value creation.

This paper develops a *Mean-Field Stackelberg Game (MFSG)* framework to model this feedback loop and characterize its equilibrium properties. The platform acts as a Stackelberg leader, choosing an algorithmic policy along two dimensions: *Retention Control* (real-time interventions that adjust the drift of user attention) and *Matching Control* (investment in recommendation quality that shifts the distribution of future content). A continuum of users, acting as rational followers, solve an optimal stopping problem against the aggregate policy. The closed-loop nature of traffic recycling enters through a non-local boundary condition: the mass of users who swipe away is reinjected into the system rather than absorbed. We calibrate the model’s behavioral parameters using KuaiRand, a large-scale randomized experiment dataset from Kuaishou, and solve the resulting coupled system via a physics-informed deep reinforcement learning algorithm.

Three analytical results emerge, each connected by a single causal chain. The first, which we term the *Dopamine Treadmill* (Proposition 4), establishes the micro-foundation: improving recommendation quality raises the expected value of the next video, which in turn raises users’ rational threshold for tolerating the current one. Better algorithms make users more impatient—not less. This mechanism is self-reinforcing inside the traffic

recycling loop: faster swiping increases the pool of videos a user evaluates per session, further elevating expectations and compressing dwell times.

The second result, the *Algorithmic Liquidity Trap* (Theorem 1), is the central contribution. We prove analytically that the equilibrium dwell time τ^* is a strictly decreasing function of the platform’s monetization ratio $\mathcal{R} = \lambda_2/\lambda_1$ (ad revenue weight relative to retention weight), and that a critical threshold $\mathcal{R}^*$ exists beyond which τ^* falls below the IAB/MRC 2-second viewability standard. The key insight, which follows from Little’s Law applied to the stationary ecosystem, is that $\tau^* = 1/\phi_{\text{swipe}}^*$: dwell time is the reciprocal of the per-capita swipe rate. Once the platform optimizes aggressively for flux, the entire effective revenue collapses—the nominal traffic surge is pure Phantom Liquidity. Crucially, this is not a coordination failure between competing platforms, but an internal contradiction within a single platform’s own unconstrained objective.

The third result characterizes the *structure* of the optimal algorithmic intervention. The platform’s equilibrium policy is non-linear: it applies peak negative stimulation (accelerating boredom) at intermediate attention states, and positive reinforcement only near the churn boundary, creating a “Trap-and-Release” cycle. This policy is locally optimal for ad extraction but constitutes a “Tragedy of the Commons” at the ecosystem level, depleting the user’s capacity for sustained attention and, through the Dopamine Treadmill, driving further compression of the threshold for future sessions.

Together, these three results reframe platform algorithm design as a *mechanism design problem*: the platform’s privately optimal unconstrained algorithm produces a socially and commercially inferior equilibrium. Constraints on dwell time—such as those embedded in completion-rate weighting, which major platforms including TikTok and Kuaishou already employ—are not regulatory impositions but economically necessary corrections to an internal market failure. Our framework provides the first theoretical foundation for why such constraints are not merely pro-social, but are required for the platform’s own long-run revenue maximization.

Contributions. This paper makes three contributions. *Theoretically*, we introduce the first continuous-time model of platform algorithm design that incorporates traffic recycling as a non-local feedback mechanism, deriving a coupled HJB/Fokker-Planck system whose equilibrium properties include the Algorithmic Liquidity Trap. *Empirically*, we provide structural calibration of the model’s behavioral parameters using large-scale randomized experiment data, grounding the theoretical predictions in physiologically realistic parameter values. *Managerially*, we characterize the precise structure of the optimal constrained algorithm, show that retention-first policies strictly dominate ad-centric policies for all possible revenue mixes between advertising and subscription, and identify the critical monetization ratio $\mathcal{R}^*$ above which platform governance interventions become economically necessary.

The remainder of this paper is organized as follows. Section 2 positions this work within three streams of literature. Section 3 develops the MFSG model. Section 4 details the structural calibration. Section 6 presents the equilibrium analysis. Section 7 concludes

with managerial and policy implications.

2 Literature Review

This paper contributes to three streams of literature: the economics of optimal stopping and consumer search, the operations management of algorithmic platforms, and mean-field game theory applied to large-scale strategic interactions. Our central contribution relative to each stream is summarized below.

2.1 Rational Attention and Optimal Stopping

The micro-foundation of our model builds upon the classical theory of optimal stopping under uncertainty. The seminal works of [Stokey \(2008\)](#) and [Peskir and Shiryaev \(2006\)](#) established the mathematical rigor for single-agent stopping problems driven by geometric Brownian motions. In the context of the digital economy, this framework has been adapted to model online search behavior. For instance, [Gershkov and Moldovanu \(2009\)](#) analyze consumer search with learning, while [Ke et al. \(2016\)](#) propose a Bayesian framework for rapid information consumption.

However, these classical models typically assume an “absorbent” exit state—once a user stops (or purchases), they leave the system. Our work extends this by introducing the concept of “*Traffic Recycling*,” where the “stopping” action (swiping) is not an exit but a transition back into the content pool. This requires a non-local boundary condition that couples the exit flux of one state to the entry flux of another, a dynamic absent in standard search theory.

2.2 Algorithmic Management of Attention in Recommender Systems

The second stream focuses on the operational design of content platforms. In the computer science domain, early works prioritized binary metrics such as Click-Through Rate (CTR). More recently, industrial frameworks like those at YouTube ([M. Chen et al., 2019](#); [Covington, Adams, & Sargin, 2016](#)) have shifted towards continuous engagement metrics (e.g., watch time) via Deep Reinforcement Learning.

A theoretical gap persists in the OM and economics literature, however, regarding the *objective function* of platform algorithms. Standard models in online advertising and revenue management treat impression flux and ad revenue as linearly related (e.g., [Goldfarb & Tucker, 2011](#); [Lobel, Patel, Vulcano, & Zhang, 2016](#))—effectively assuming that attention is a divisible commodity whose value scales proportionally with throughput. This assumption is empirically violated by the IAB/MRC viewability standard, under which an impression has positive value only if dwell time exceeds two seconds. The step-function structure of the revenue-dwell relationship means that the distinction between *nominal flux* (server-side impressions) and *effective economic flow* (billable impressions) is not merely

quantitative but categorical. No existing OM model captures this discontinuity or its implications for algorithm design.

Our paper fills this gap. By embedding the IAB threshold into the platform’s equilibrium analysis, we show that the “Linear Flux Hypothesis” is not merely imprecise but misleading: it predicts a monotone relationship between algorithmic aggressiveness and revenue, while the true equilibrium relationship is non-monotone and admits a catastrophic collapse beyond a critical monetization ratio. On the demand side, [Bao \(2025\)](#) provides complementary micro-evidence that short-form content systematically exacerbates user self-control problems; our model endogenizes the platform mechanism that drives this behavior from the supply side.

2.3 Mean-Field Stackelberg Games in Platform Economies

Finally, we contribute to the emerging literature on mean-field methods applied to platform economics. This literature spans two related but distinct frameworks that we now carefully position.

Standard Mean-Field Games (MFG) in the sense of [Lasry and Lions \(2007\)](#) model symmetric agents who each optimize their own objective in Nash equilibrium with the aggregate distribution. Users interact *directly* through the mean-field—each agent’s optimal action depends on the distribution of all others’ states. Applied to platform settings, [Iyer, Johari, and Sundararajan \(2014\)](#) and [Yang et al. \(2018\)](#) use MFG-type approximations for large-scale information systems.

Mean-Field Control (MFC), by contrast, places a social planner who maximizes a collective welfare functional over the entire population, internalizing the aggregate effect ([Carmona, Delarue, et al., 2018](#)). Neither framing fits our setting: users are autonomous rational agents (not socially planned), yet they do not play strategically *against each other*—each user independently solves an optimal stopping problem, taking the platform’s algorithmic policy as given.

Our framework is a *Mean-Field Stackelberg Game (MFSG)*: a hierarchical structure in which the *platform* (Stackelberg leader) commits to an algorithmic policy, and a *continuum of rational users* (followers) best-respond independently. The “game” is between the platform and the aggregate mass of users, not among users themselves. Traditional heterogeneous-agent frameworks face the “curse of dimensionality” as $N \rightarrow \infty$; the mean-field approximation replaces the N -dimensional joint distribution with the tractable population density $m(t, z)$ ([Lasry & Lions, 2007](#)).

Three features of our model make the mean-field population structure strictly necessary—and cannot be collapsed to a representative-agent control problem: (i) users are heterogeneous in their current attention state $z \in \Omega$, so the platform’s optimal control $a^*(z)$ must be a function of z , not a scalar; (ii) aggregate outcomes (N^*, Φ^*, τ^*) are moments of $m(t, z)$ and depend on its full shape; (iii) the non-local recycling injects swiped users from the exit boundary back into the interior via $g(z; u)$, creating distributional heterogeneity that a single representative state cannot capture.

Our technical novelty within this MFSG class is the *non-local Fokker-Planck structure*: unlike standard MFGs where population mass vanishes at the boundary, our model is “re-entrant”—the outflow at b^* is immediately reinjected into the content pool. This closed-loop recycling, absent from both classical MFG and MFC, is what generates the Phantom Liquidity phenomenon.

Our contribution sits at the intersection of all three streams: we model the user’s optimal stopping decision (Stream 1), study the platform’s algorithmic design problem subject to an IAB/MRC revenue discontinuity (Stream 2), and characterize the resulting large-population equilibrium via the mean-field approach (Stream 3). The interplay among these three components generates the Algorithmic Liquidity Trap—a phenomenon that is invisible to any single-stream analysis.

3 The Model

We formulate the interaction between the short-video platform and its massive user base as a Mean-Field Stackelberg Game. The platform (Leader) commits to an algorithmic policy to maximize long-term rewards, while individual users (Followers) observe the ecosystem’s state and optimally decide when to “swipe” to the next video.

3.1 Empirical Motivation: Stylized Facts

Before formalizing the stochastic dynamics, we summarize three critical stylized patterns observed in the high-frequency interaction data (KuaiRand). These patterns serve as the empirical micro-foundations for our specific design of the attention dynamics and traffic recycling mechanism.

- **Pattern 1: Rapid Initial Filtering (The “Golden Seconds”).** The empirical distribution of dwell times reveals a sharp density spike in the interval $[0, 2s]$, indicating that users make rapid, reflex-driven rejection decisions. This motivates our use of an **Absorbing Boundary** at a lower threshold b^* rather than a purely duration-based utility model.
- **Pattern 2: Heavy-Tailed Engagement (The “Gambling” Effect).** Despite the rapid decay, the survival function exhibits a fat tail (high kurtosis). This suggests that attention is driven by high volatility (σ), validating the **Geometric Brownian Motion** assumption where users chase stochastic rewards.
- **Pattern 3: Closed-Loop Circulation (Traffic Recycling).** Analysis of session logs shows that a “swipe” event rarely terminates the session ($P(\text{Exit}) \approx 1.25\%$). Instead, it triggers an immediate re-entry. This necessitates a **Non-Local PDE framework** where the outflow mass is reinjected into the system, rather than a standard optimal stopping model with permanent exit.

3.2 A Motivating Two-Period Model

Before developing the continuous-time framework, we expose the core tension in a minimal two-period discrete model. This stripped-down setting delivers a closed-form characterization of the platform’s trade-off and provides an analytic benchmark for the richer results that follow.

The three stylized patterns identified above directly shape this model’s structure. Pattern 3 (traffic recycling, $P(\text{Exit}) \approx 1.25\%$) motivates the recycling term: swiped-away users re-enter rather than permanently exit. Pattern 1 (rapid filtering) motivates the corner solution: under flux-maximizing incentives the algorithm drives retention toward zero, collapsing dwell time to a physiological floor. The full continuous-time framework (Section 3.3) embeds all three patterns simultaneously, and the structural parameters $(\mu, \sigma, \varepsilon)$ are calibrated to KuaiRand data in Section 4.

Setup. Let $p \in [0, 1]$ denote the *retention probability*: the probability that a user, after watching one video, decides to stay and watch a second video (rather than swipe away). Given a unit mass of users in period 1, the mass present in period 2 is p . Users who swipe away are immediately recycled by the algorithm and re-enter, contributing $(1 - p)$ to period-2 exposure. Thus:

- **Total engagement time** (proportional to retention): $T(p) = 1 + p =$ period-1 mass plus returning users.
- **Total flux** (swipe events, proportional to ad impressions between clips): $\Phi(p) = (1 - p) \cdot 2 + p \cdot 1$. More precisely, with two periods and traffic recycling, one can show:

For concreteness, suppose each user generates 1 ad impression per swipe event, and retained users swipe once in period 2 while recycled users swipe twice (once to exit, once to re-enter). Then:¹

$$T(p) = p(3 - p), \quad \Phi(p) = (1 - p)(2 - p). \quad (1)$$

The platform maximizes a weighted objective over engagement and ad revenue:

$$V(p) = \lambda_1 T(p) + \lambda_2 \Phi(p) = \lambda_1 p(3 - p) + \lambda_2 (1 - p)(2 - p), \quad (2)$$

where $\lambda_1 > 0$ weights retention-driven revenue (subscriptions, long-term loyalty) and $\lambda_2 > 0$ weights flux-driven revenue (advertising).

¹These quadratic forms satisfy the boundary conditions $T(0) = 0, T(1) = 2, \Phi(0) = 2, \Phi(1) = 0$, consistent with the verbal description above. Their derivatives are $T'(p) = 3 - 2p > 0$ and $\Phi'(p) = -(3 - 2p) < 0$ for all $p \in [0, 1]$, so T is strictly increasing in retention while Φ is strictly decreasing—the fundamental trade-off. Crucially, $T'(p) \cdot \Phi'(p) < 0$ for all interior p , which implies the FOC $dV/dp = 0$ has no interior solution and Proposition 1 (corner solution at $R^* = 1$) holds for *any* differentiable T, Φ satisfying this sign condition; the quadratic specification is illustrative.

Optimal Policy and the Critical Ratio. Taking the derivative with respect to p :

$$\frac{dV}{dp} = \lambda_1(3 - 2p) - \lambda_2(3 - 2p) = (\lambda_1 - \lambda_2)(3 - 2p). \quad (3)$$

Since $3 - 2p > 0$ for all $p \in [0, 1]$, the sign of dV/dp is determined entirely by $\lambda_1 - \lambda_2$. Define the *revenue ratio* $R \equiv \lambda_2/\lambda_1$. Then:

Proposition 1 (Two-Period Algorithmic Liquidity Trap). *The platform’s optimal retention probability is a corner solution:*

$$p^* = \begin{cases} 1 & \text{if } R < 1 \quad (\text{retention-dominant regime}), \\ 0 & \text{if } R > 1 \quad (\text{flux-dominant regime}). \end{cases}$$

At the critical ratio $R^* = 1$, the objective is flat and any $p \in [0, 1]$ is optimal.

The Phantom Liquidity Collapse. When $R > 1$, the platform drives $p^* \rightarrow 0$, maximizing swipe flux. Substituting into Little’s Law, the mean dwell time satisfies:

$$\tau^* = \frac{N^*}{\Phi^*} = \frac{T(0)}{\Phi(0)} = \frac{0}{2} = 0. \quad (4)$$

Thus $\tau^* < 2$ s—below the empirical threshold for meaningful cognitive engagement (Pattern 1 above). Despite generating high nominal flux $\Phi^* = 2$, ad impressions occur so rapidly that conversion rates collapse. We term this the *Algorithmic Liquidity Trap*: the platform optimally self-destructs its own engagement quality.

What Does the Continuous-Time Framework Add Beyond the Two-Period Model? This is a question a careful reader will rightfully ask, and it deserves a direct answer. The two-period model establishes the *existence* of the trap: it shows that a critical monetization ratio $R^* = 1$ separates healthy from pathological regimes. But it can say nothing about four questions that are central to the paper’s managerial and regulatory contributions.

First, optimal policy structure. Proposition 1 only tells us the platform should set $p^* = 0$ in the flux-dominant regime. It cannot characterize *how* the algorithm achieves this in practice. The continuous-time framework reveals the specific shape of the optimal control field $a^*(z)$: strong negative drift at mid-range attention states (“Planned Obsolescence”) combined with a thin protective band near the exit boundary (“Safety Net”). This Trap-and-Release architecture is not a corollary of the two-period model—it is a novel structural finding that can be directly searched for in logged behavioral data and used as a regulatory diagnostic.

Second, the Dopamine Treadmill. The two-period model treats content quality as fixed. The continuous-time framework endogenizes the user’s outside option $K(u)$ as a functional of the quality distribution $g(z; u)$, yielding Proposition 4: improving content quality *raises* the reservation threshold b^* , paradoxically compressing dwell times and accelerating the recycling loop. This mechanism—invisible in any two-period or static model—has

direct implications for content investment strategy and explains the structural difficulty of quality-driven escapes from the trap.

Third, quantitative calibration against viewability standards. The two-period model predicts $\tau^* = 0$ at the corner solution, which is economically correct but managerially useless for threshold-setting. By calibrating $(\mu, \sigma, \varepsilon)$ to KuaiRand data, the continuous-time framework generates specific predictions: $\tau^* \approx 24$ s in the Baseline regime (92% of flux billable under IAB/MRC) versus $\tau^* \approx 0.10$ s in the Clickbait regime ($\approx 10^{-9}$ billable fraction). These numbers locate both regimes precisely on the viewability schedule and allow the dominance result to be stated not just in sign but in magnitude.

Fourth, equilibrium multiplicity and fragility. Because the two-period model collapses to a corner solution at $R^* = 1$, it predicts a single equilibrium on each side of the bifurcation. The continuous-time framework reveals that near $\mathcal{R} = 1$ the equilibrium manifold supports *multiple* coexisting fixed points with different basin sizes—reflected empirically in the asymmetric DRL convergence rates (9/10 for Baseline, 7/10 for Clickbait) and the non-trivial cross-seed dispersion in b^* . This fragility result carries an additional policy implication that the two-period model cannot deliver: the clickbait ecosystem is structurally fragile to perturbations, making small governance interventions disproportionately effective near the critical ratio.

In summary, the continuous-time MFSG is not simply a technical embellishment of the two-period intuition. It is the minimal framework needed to answer *how* the trap works, *why* quality investment fails, *exactly where* the ecosystem sits relative to the IAB/MRC threshold, and *how fragile* the pathological equilibrium is.

3.3 The Micro-Ecosystem: Attention Dynamics and Retention Control

Building on these stylized patterns, we now formally define the user’s attention process.

Scope of Analysis: The Demand-Side Loop. To isolate the mechanism of the “Algorithmic Liquidity Trap,” we focus our analysis on the closed-loop interaction between the *Platform’s Algorithmic Policy* and the *User’s Attention Dynamics*. We treat the content supply side (creators) as an instrument controllable by the platform via the matching variable $u(t)$, which shifts the quality distribution $g(z; u)$. While we acknowledge that creators inevitably react to algorithmic incentives in the long run (a supply-side response), explicitly modeling the creator game would introduce significant tractability issues without altering the fundamental existence of the liquidity trap driven by user physiology. We thus abstract away strategic creator behavior to focus on the immediate feedback loop between algorithmic acceleration and cognitive capacity.

Assumption (The Infinite Feed Paradigm). We restrict our analysis to the *Feed-based Infinite Scroll* mechanism (e.g., TikTok’s “For You” page), characterized by passive consumption and high-frequency switching. We distinguish this from *search-based* or *long-form* consumption (e.g., Netflix), which follows different decision dynamics.

The Algorithmic Liquidity Trap — How the Feedback Loop Works

Zhu & Zhu (2026) · Mean-Field Stackelberg Game with Traffic Recycling

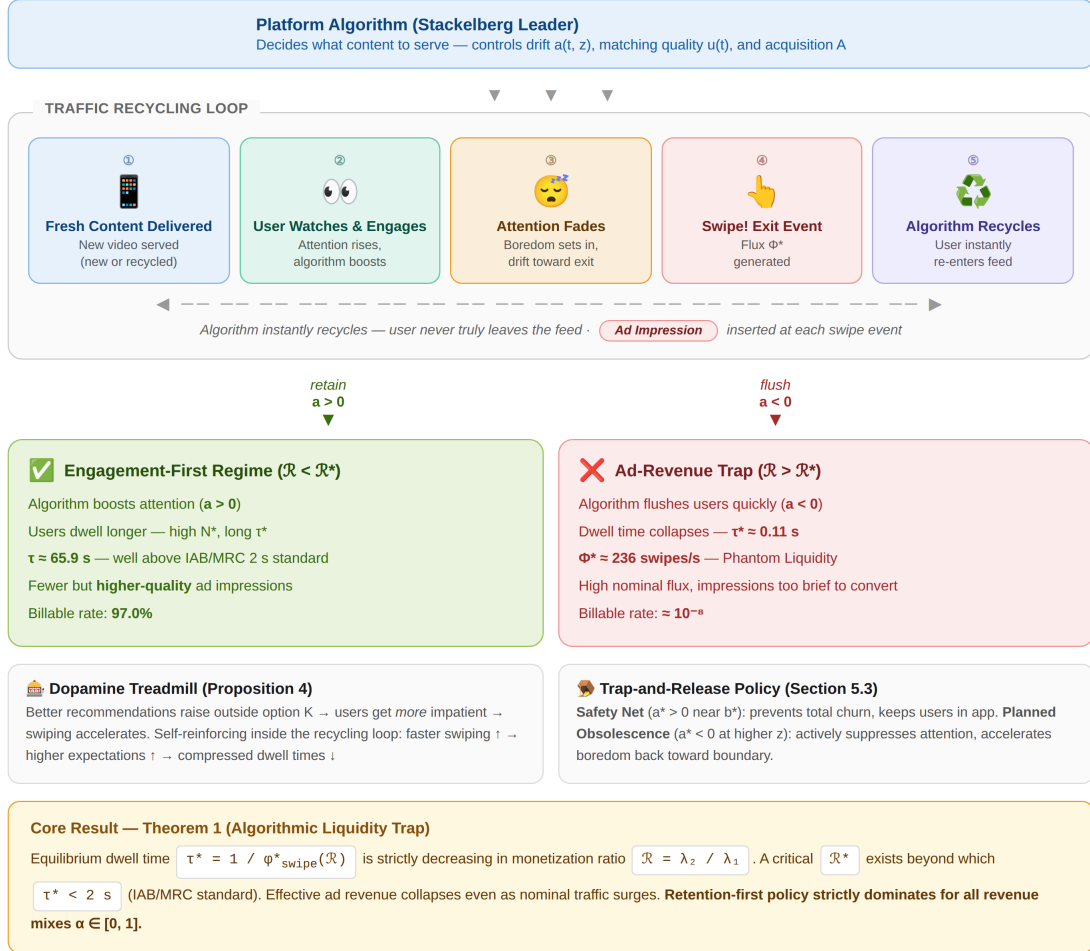


Figure 1: Framework overview of the Mean-Field Stackelberg Game with Traffic Recycling. *Top*: The platform (Stackelberg leader) controls algorithmic drift $a(t, z)$, matching quality $u(t)$, and acquisition A . *Middle*: Users cycle through five stages — content delivery, engagement, attention decay, a swipe event (generating flux Φ^*), and immediate algorithmic recycling; an ad impression is inserted at each swipe. *Bottom left*: In the engagement-first regime ($\mathcal{R} < \mathcal{R}^*$), the algorithm retains users, yielding high N^* and long dwell time τ^* . *Bottom right*: In the algorithmic liquidity trap ($\mathcal{R} > \mathcal{R}^*$), the algorithm flushes users rapidly; τ^* collapses toward zero and effective ad revenue vanishes despite high nominal flux.

The dynamics of the user population are driven by the interplay between continuous engagement (viewing) and discrete resampling events (swiping). We first describe the continuous phase.

Controlled Attention Process. Consider a continuum of ex-ante homogeneous users indexed by $i \in [0, N]$. Let $Z_t^i \in \mathbb{R}_+$ denote the instantaneous attention level (or physiological arousal) of user i at time t during the viewing of a single video. We model Z_t^i as a controlled Geometric Brownian Motion (GBM):

$$dZ_t^i = (\mu + a(t, Z_t^i)) Z_t^i dt + \sigma Z_t^i dW_t^i, \quad (5)$$

where W_t^i are independent standard Brownian motions capturing the intrinsic volatility of content stimuli.

- **Natural Decay** ($\mu < 0$): The parameter μ represents the baseline rate of attention decay (i.e., the onset of boredom). In the absence of stimuli, user attention naturally drifts toward zero, reflecting the entropy of the swiping mechanism.
- **Algorithmic Modulation** ($a(t, z)$): The control variable $a(t, z) \in [a_{\min}, a_{\max}]$ (where $a_{\min} < 0$) represents the platform’s *intervention intensity*. Unlike traditional editorial editing, this corresponds to bidirectional *Sensory Modulation*:
 - **Positive Stimulation** ($a > 0$): Using real-time features (e.g., visual hooks, intense audio cues) to artificially inject a positive drift that *counteracts* natural decay (Retention).
 - **Negative Nudging** ($a < 0$): Strategically reducing stimuli or withholding gratification to *accelerate* decay. This allows the algorithm to proactively “flush” users out of low-yield states to trigger a new swipe (Impression), serving as a micro-foundation for the Algorithmic Liquidity Trap.

Assumption 1 (Natural Decay). The drift parameter satisfies $\mu < 0$, so that user engagement decays naturally in the absence of platform intervention.

We denote the effective drift as $\tilde{\mu}(t, z) = \mu + a(t, z)$. This SDE governs the user’s path *only* until an endogenous stopping time τ^* . Upon hitting the churn boundary $b^*(t)$, the user exits the current path and triggers a “swipe” event, governed by the traffic recycling mechanism.

Modeling Remark: From Agents to Fields. In defining the macroscopic evolution, we adopt a “top-down” Mean-Field approach. While methods like heterogeneous-agent models (Aiyagari, 1995; Nuño & Moll, 2018) are valuable in macroeconomics, they often rely on static or finite-horizon equilibrium concepts. Given our focus on the *velocity* of user flows and the *recycling* dynamics, we model the aggregate state directly via a Mean-Field PDE. Crucially, our ecosystem is an *open system with internal recycling*, governed by a *Non-Local*

Fokker-Planck Equation. This structure allows us to capture the complex feedback between the churn boundary b^* and the re-entry distribution $g(z)$ —a necessary feature to model the “Algorithmic Liquidity Trap.”

3.4 The User’s Problem: Optimal Stopping with Endogenous Thresholds

Given the controlled attention dynamics described in Equation (5), the user solves an optimal stopping problem to determine the precise moment to “swipe” away from the current video.

Value Function and the HJB Equation. The user’s objective is to maximize total discounted utility. The value function $v(t, z)$ is defined as the supremum over all admissible stopping times τ :

$$v(t, z) = \sup_{t \leq \tau < \infty} \mathbb{E} \left[\int_t^\tau e^{-\rho(s-t)} u_{\text{util}}(Z_s) ds + e^{-\rho(\tau-t)} (K(\tau, u(\tau)) - C) \mid Z_t = z \right]. \quad (6)$$

We employ a CRRA utility function $u_{\text{util}}(z) = \frac{z^{1-\gamma}}{1-\gamma}$ (with $\gamma > 1$). The term C represents the cognitive switching cost.

Crucially, the payoff upon switching, $K(t, u)$, is not fixed but depends on the platform’s concurrent *Matching Control* $u(t)$. This constitutes a *Variational Inequality* (VI) for the value function:

$$\min \left\{ \rho v - \frac{\partial v}{\partial t} - \mathcal{L}^a v - u_{\text{util}}(z), v(t, z) - (K(t, u(t)) - C) \right\} = 0, \quad (7)$$

where $\mathcal{L}^a$ is the generator of the controlled diffusion:

$$\mathcal{L}^a v = (\mu + a(t, z)) z \frac{\partial v}{\partial z} + \frac{1}{2} \sigma^2 z^2 \frac{\partial^2 v}{\partial z^2}. \quad (8)$$

The *optimal stopping boundary* (or *Swiping Threshold*) $b^*(t)$ is the free boundary separating the continuation region from the stopping region:

$$b^*(t) = \inf \{ z \in \Omega \mid v(t, z) > K(t, u(t)) - C \}. \quad (9)$$

Equivalently, under the threshold structure, $b^*(t)$ is characterized by the *value-matching* condition

$$v(t, b^*(t)) = K(t, u(t)) - C. \quad (10)$$

Users with attention $z \leq b^*(t)$ immediately execute a “swipe”.

Boundary Conditions (Value-Matching and Smooth-Fit). The value-matching condition requires $v(t, b^*(t)) = K(t, u(t)) - C$ at the boundary. The smooth-pasting condition requires continuity of the spatial derivative: $\partial_z v(t, b^*(t)) = \partial_z [K(t, u(t)) - C] = 0$, since

$K(t, u(t)) - C$ is independent of z . That is,

$$\frac{\partial v}{\partial z}(t, b^*(t)) = 0. \quad (11)$$

At the artificial truncation boundaries $z_{\min}, z_{\max}$ (used in our numerical solver), we impose reflecting (Neumann) conditions consistent with the conservative fixed-domain scheme:

$$\frac{\partial v}{\partial z}(t, z_{\min}) = 0, \quad \frac{\partial v}{\partial z}(t, z_{\max}) = 0, \quad (12)$$

together with a standard growth/transversality condition ensuring finiteness of v under discounting. *Remark (Numerical truncation)*: these boundary conditions are imposed on the truncated computational domain for numerical stability; in our experiments we choose $z_{\max}$ sufficiently large (and $z_{\min}$ sufficiently small) and confirm that the computed policy and threshold are insensitive to further enlarging the domain.

The Endogenous Outside Option (Coupling Variable). The variable $K(t, u)$, which serves as the boundary condition for the user’s problem, represents the expected value of the “next video”. In our model, this creates a *Fixed-Point problem*: the value of the outside option depends on the value function itself. Specifically, K is determined by the platform’s investment $u(t)$, which shapes the content quality distribution $g(z; u)$:

$$K(t, u(t)) = \int_{z_{\min}}^{z_{\max}} v(t, z') g(z'; u(t)) dz'. \quad (13)$$

Remark (Implicit Fixed-Point Structure). Since $v(t, z') = K(t, u(t)) - C$ for all $z' \leq b^*(t)$ (by value-matching), Equation (13) is an *implicit* equation in K . Letting $G(t) \equiv \int_{z_{\min}}^{b^*(t)} g(z'; u) dz'$ denote the probability that the next video lands in the stopping region, one can verify that $K = G(K - C) + \int_{b^*(t)}^{z_{\max}} v g dz'$, i.e., $K = \frac{-G C + \int_{b^*(t)}^{z_{\max}} v g dz'}{1 - G}$. In our numerical solver we treat this as a fixed-point iteration: given a candidate K , we solve the QVI for v , update K via (13), and iterate to convergence.

This equation highlights the *Dopamine Treadmill* effect: An increase in platform matching investment $u(t)$ shifts $g(z; u)$ towards higher quality. This raises the expected switching value K , which in turn raises the user’s stopping threshold $b^*(t)$. Paradoxically, better recommendation algorithms may make users *more* impatient, as the opportunity cost of watching mediocre content increases. This mechanism mathematically drives the acceleration of swiping velocity, serving as the micro-foundation for the “High-Velocity Saturation” regime observed in our results.

Theoretical Well-posedness

Before characterizing the equilibrium, we establish the theoretical well-posedness of the user’s sub-problem defined by the QVI in (7).

Proposition 2 (Well-posedness of the User QVI). *Given a fixed platform control policy $\{a(t, z), u(t)\}$*

and a fixed outside option process $K(t, u(t))$, the QVI in (7) admits a unique viscosity solution $v(t, z)$ under standard boundedness and discounting conditions (Fleming & Soner, 2006).

Moreover, suppose the re-entry distribution $g(\cdot; u(t))$ is a nonnegative density on $[z_{\min}, z_{\max}]$ with $\int g \, dz = 1$ and $\|g\|_{\infty} < \infty$, and we work on the truncated state domain with discounting $\rho > 0$ so that v is finite. Then the coupling (13) defines a well-posed fixed-point problem in K . In our numerical implementation we solve it via a monotone (relaxed) value-iteration scheme and verify convergence of K in sup norm.

Proposition 3 (Threshold Structure). *For the CRRA utility function with $\gamma > 1$, the value function $v(t, z)$ is monotonically increasing in z . Consequently, the optimal stopping region is connected and characterized by a unique threshold $b^*(t)$:*

$$\mathcal{S}_t = [z_{\min}, b^*(t)]. \quad (14)$$

Proof. The result follows from the monotonicity of the generator for one-dimensional diffusions. Detailed proof is provided in Appendix A.1.

This structural property is computationally significant: it allows us to search for a single scalar $b^*(t)$ (e.g., via bisection) rather than evaluating the stopping decision for every grid point, reducing the complexity from $O(N_z)$ to $O(\log N_z)$.

Proposition 4 (Dopamine Treadmill). *Let $g(\cdot; u')$ first-order stochastically dominate $g(\cdot; u)$, i.e., the platform's matching investment $u' > u$ shifts the content quality distribution toward higher values. Then the equilibrium stopping threshold satisfies $b^*(u') > b^*(u)$: a higher-quality content pool makes users strictly more impatient.*

Proof. By Proposition 2, $v(t, z)$ is strictly increasing in z . First-order stochastic dominance of $g(\cdot; u')$ over $g(\cdot; u)$ therefore implies

$$K(u') = \int v(z') g(z'; u') \, dz' > \int v(z') g(z'; u) \, dz' = K(u). \quad (15)$$

At the original threshold $b^*(u)$, value-matching gives $v(b^*(u)) = K(u) - C < K(u') - C$. Hence the outside option $K(u') - C$ now strictly exceeds the continuation value at $b^*(u)$, meaning that users strictly prefer to stop at $z = b^*(u)$ when the outside option is $K(u')$. By the definition of the threshold as the infimum of the stopping region, $b^*(u') \geq b^*(u)$; strict first-order dominance ensures the inequality is strict. $\square$

Economic interpretation. Improving the recommendation algorithm raises the expected value of the “next video” K , increasing the opportunity cost of any given video. Users rationally demand a higher current attention level to justify staying, so they swipe sooner. This explains the structural difficulty of sustaining long-form content in a pure feed architecture: algorithmic improvements that benefit the *average* content quality paradoxically render *individual* videos less sticky.

3.5 The Macro-Dynamics: Non-Local Fokker-Planck Equation

As the population size $N \rightarrow \infty$, the aggregate behavior of users is described by the population density $f(t, z)$ (an *unnormalized* measure density). The evolution of $f(t, z)$ follows mass balance, adjusted for inflows (acquisition and recycling) and outflows (true churn).

Derivation of the Non-Local PDE. Standard arguments involving Itô’s formula and integration by parts (see [Gardiner et al., 2004](#)) yield the differential operator $\mathcal{L}^*$ adjoint to the generator $\mathcal{L}^a$. However, unlike standard Fokker-Planck equations that assume mass conservation, our ecosystem acts as an open system with internal loops.

Physical domain and absorption. Conceptually, the density evolves on the time-dependent physical domain $\Omega_t = [b^*(t), z_{\max}]$ (users who hit $b^*(t)$ stop watching the current video). On Ω_t we impose the absorbing boundary condition $f(t, b^*(t)) = 0$ and a reflecting boundary at $z_{\max}$.

Non-local sources and units. The re-entry kernel $g(\cdot; u(t))$ is a probability density on $[z_{\min}, z_{\max}]$ with $\int g(z; u) dz = 1$ (units $1/z$). Therefore, $(1 - \epsilon)\Phi_{\text{swipe}}(t)g(z; u(t))$ has units “users/(time $\cdot$ z)”, matching $\partial_t f$.

Under this physical interpretation, the density satisfies:

$$\frac{\partial f}{\partial t}(t, z) = \underbrace{\mathcal{L}^* f(t, z)}_{\text{Gaze Dynamics}} + \underbrace{A(t)\delta(z - z_p)}_{\text{External Acquisition}} + \underbrace{(1 - \epsilon)\Phi_{\text{swipe}}(t)g(z; u(t))}_{\text{Traffic Recycling}}, \quad z \in \Omega_t. \quad (16)$$

Remark (Decoupling of Φ_{swipe}). Although $\Phi_{\text{swipe}}(t)$ appears on the right-hand side of (16), it is itself a functional of f . In the continuous-time limit, the absorption at $b^*(t)$ is expressed via the probability current defined in (19): $\Phi_{\text{swipe}}(t) = |J(t, b^*(t))|$, which depends only on f and $\partial_z f$ at the boundary. Substituting this expression into (16) fully decouples the equation; the fixed-domain “cut” scheme in our numerical implementation approximates this boundary current by mass truncation (Section B).

Mass balance implied by absorption. Integrating (16) over Ω_t and using the reflecting boundary at $z_{\max}$ yields

$$\frac{d}{dt} \int_{\Omega_t} f(t, z) dz = A(t) - \epsilon \Phi_{\text{swipe}}(t), \quad (17)$$

where the loss term is the non-recycled fraction ϵ of the swipe flux (true churn).

Fixed-domain implementation (cut). For numerical stability, we implement a conservative fixed-domain scheme on $\Omega = [z_{\min}, z_{\max}]$ and enforce the absorption at $b^*(t)$ via a projection (cut) after each implicit PDE step; this cut removes exactly the swipe mass, recycles a fraction $(1 - \epsilon)$ via g , and loses a fraction ϵ , matching (17) discretely.

where the adjoint operator is:

$$\mathcal{L}^* f = -\frac{\partial}{\partial z} [(\mu + a(t, z))zf] + \frac{1}{2} \frac{\partial^2}{\partial z^2} [\sigma^2 z^2 f]. \quad (18)$$

We denote the associated probability current by

$$J(t, z) \triangleq (\mu + a(t, z))zf(t, z) - \frac{1}{2} \frac{\partial}{\partial z} (\sigma^2 z^2 f(t, z)). \quad (19)$$

Interpretation of Source Terms. Equation (16) distinguishes our model from standard MFG frameworks through two distinct source mechanisms:

1. **External Acquisition (Dirac Source):** The term $A(t)\delta(z - z_p)$ represents the injection of new users. The control variable $A(t)$ corresponds to marketing spend (akin to the Bass, 1969 diffusion model), while the Dirac delta $\delta(z - z_p)$ signifies a standardized onboarding process.
2. **Traffic Recycling (The Feedback Loop):** The term $(1 - \epsilon)\Phi_{swipe}(t)g(z; u(t))$ captures the closed-loop nature of short-video platforms. When a user swipes, a fraction $(1 - \epsilon)$ immediately re-enters the cycle. *Theoretical Note:* This non-local coupling is the mathematical root of the “Algorithmic Liquidity Trap.” Since the platform can boost population density $f(t, z)$ by accelerating the recycling flux $\Phi_{swipe}(t)$, unconstrained optimization algorithms may aggressively drive users to swipe (increasing Φ) rather than extending their gaze (retention), inadvertently creating a high-velocity, low-value ecosystem.

Boundary Conditions and Flux Dynamics. To align with our conservative “solver-in-the-loop” implementation on a fixed computational container, we evolve $f(t, z)$ on the static domain $\Omega = [z_{\min}, z_{\max}]$ subject to *zero-flux (reflecting)* boundary conditions at both ends:

$$f(0, z) = f_0(z), \quad (\text{Initial Distribution}) \quad (20)$$

$$J(t, z_{\min}) = 0, \quad J(t, z_{\max}) = 0. \quad (\text{Zero-Flux Boundaries}) \quad (21)$$

The reflecting boundaries prevent artificial mass leakage through the computational container during each implicit time step.

The *absorbing* nature of the endogenous stopping threshold $b^*(t)$ is then enforced via a *projection (cut)* condition applied after each PDE update:

$$f(t, z) = 0, \quad \forall z < b^*(t), \quad (22)$$

so that any mass that has entered the stopping region is strictly removed and counted as swipe flux.

$$\Phi_{swipe}(t_n) \triangleq \frac{1}{\Delta t} \int_{z_{\min}}^{b^*(t_n)} f^*(t_n, z) dz. \quad (23)$$

where $f^*(t_n, \cdot)$ denotes the post-implicit (pre-cut) density in our conservative time-stepping scheme. This discrete integral formulation is numerically robust and guarantees exact consistency between the computed swipe flux and the mass removed by the projection step.

In the smooth continuous-time limit, one may relate the swipe flux to the probability current at the threshold via $\Phi_{swipe}(t) \approx -J(t, b^*(t))$. *Important clarification:* throughout the paper, our *numerical* swipe flux is computed by the conservative cut-based mass integration in Eq. (23) (mass removed by the projection step divided by Δt). The current-based identity is only a continuous-limit heuristic; with a time-varying free boundary $b^*(t)$, an exact flux-current equality generally requires additional moving-boundary correction terms.

This flux $\Phi_{swipe}(t)$ is the coupling variable that closes the feedback loop. In the platform’s optimization problem (defined next), this flux represents the *supply of ad inventory*. The conflict between maximizing this inventory supply (Φ) and maintaining user retention (τ) constitutes the central tension of our model.

Remark 1 (Mathematical Well-posedness). The inclusion of source terms in Fokker-Planck equations is well-grounded. Zhu (2022) establishes the existence of weak solutions for kinetic FPEs with source terms, showing that the solution norm depends on the source norm. In the context of MFG, Claisse, Ren, and Tan (2023) derive similar equations where state-dependent birth/death rates lead to solutions that are finite measures rather than probability measures—a property that perfectly matches our non-conservative user population dynamics.

Remark 2 (Numerical Implementation). While the Dirac delta $\delta(z - z_p)$ is theoretically convenient, for numerical implementation (Section 4), we approximate it using a smooth Gaussian kernel to prevent Gibbs phenomena and ensure spectral stability.

Remark 3 (Distinction from Closed Queueing Networks). It is worth noting that the “Traffic Recycling” topology bears a structural resemblance to Closed Queueing Networks (CQN) or Gordon-Newell networks (Gordon & Newell, 1967). However, our framework differs fundamentally in its *control objective*. In classical CQN, the service mechanism is typically passive, and the objective is often to manage congestion or maximize throughput given fixed service requirements. In our Short-Video Economy, the “service time” (dwell time) is *strategically endogenous*. The platform (server) actively intervenes via $a(t, z)$ to manipulate the rate of recycling. The innovation lies not just in the closed loop itself, but in the *Stackelberg gaming* of this loop: the platform optimizes the *velocity* of circulation against the user’s physiological patience, creating the unique “Algorithmic Liquidity Trap” phenomenon that standard queueing models do not capture.

3.6 The Leader’s Problem: The Platform’s Optimal Control in the MFSG

Remark 4 (MFSG vs. MFG vs. MFC). The framework in this paper is a *Mean-Field Stackelberg Game* (MFSG), not a standard Mean-Field Game (MFG) or Mean-Field Control (MFC) problem. In a standard MFG, a continuum of *symmetric* agents simultaneously optimize, producing a Nash equilibrium in which no individual agent interacts directly with others but only through aggregate statistics. In MFC (the social-planner variant), a single central controller maximizes the welfare of the whole population. Our setup is fundamentally different: the platform is a *hierarchical Stackelberg leader* that commits to a control policy

$(a(t, z), u(t), A(t))$ before users respond, while the continuum of heterogeneous users solve individual stopping problems that depend on the announced policy. There is no Nash interaction among users; the mean-field structure arises because the platform's revenue depends on the *distribution* $m(t, z)$ of user engagement, not on any single user's action. This hierarchical, distributional structure is the defining feature of MFSG and distinguishes our model from both classical MFG and MFC.

In the standard Mean-Field Game literature, the principal often maximizes a functional of the population state alone (e.g., total density). However, as identified in the audit of short-video economics, a model focusing solely on total attention time ($\int z f$) mimics a subscription-based model (like Netflix) but fails to capture the ad-driven revenue model of platforms like TikTok or Reels. In these ecosystems, revenue is generated not just by “gazing” (retention) but significantly by “swiping” (impression generation).

To resolve the “Stalemate of Perfection” paradox—where optimal retention implies zero ad inventory—we reconstruct the platform's objective and control mechanisms. The platform acts as a Stackelberg leader, optimizing a vector of controls to balance retention intensity against the need for traffic circulation (swiping).

3.6.1 Reformulation: Decoupling Scale and Structure

To explicitly separate the economic effects of market growth (Scale) from user engagement quality (Structure), we perform a change of variables on the state space. We decompose the unnormalized population density $f(t, z) \in \mathcal{M}^+(\Omega)$ into two orthogonal components: the total population mass $N(t) \in \mathbb{R}_+$ and the probability density function $m(t, z) \in \mathcal{P}(\Omega)$.

The relationship is defined as:

$$\begin{aligned} f(t, z) &= N(t) \cdot m(t, z), \\ \text{where } N(t) &= \int_{\Omega} f(t, z) \, dz, \\ \text{and } \int_{\Omega} m(t, z) \, dz &= 1. \end{aligned} \tag{24}$$

Consequently, the platform's value functional is redefined as $\mathcal{W}(f) \equiv V(N, m)$. This decomposition allows us to distinguish the platform's two strategic imperatives: managing the *market size* (via acquisition A and churn ϵ) and optimizing the *attention quality* (via retention a and matching u).

3.6.2 The Reconstructed Objective Functional and Constraints

The platform seeks to maximize the total discounted net profit over an infinite horizon. The objective functional $J(a, u, A)$, expressed in terms of the decomposed variables, combines retention value, impression revenue, and operational costs:

$$\begin{aligned}
\max_{a,u,A} J = \int_0^\infty e^{-rt} & \left[\underbrace{N(t) \int_{b^*(t)}^{z_{\max}} \left(\lambda_1 z - \frac{\xi_1}{2} a^2 \right) m \, dz}_{\text{Net Retention Value}} \right. \\
& + \underbrace{N(t) \lambda_2 \phi_{swipe}(m) - \frac{\xi_2}{2} u(t)^2}_{\text{Net Impression Value}} \\
& \left. - \underbrace{\Psi(A(t))}_{\text{Acq. Cost}} \right] dt
\end{aligned} \tag{25}$$

Economic Interpretation of the Objective: Equation (25) encapsulates the fundamental tension of the attention economy:

- **The Editor's Incentive** ($\lambda_1 z$): Rewards prolonged engagement depth (Duration).
- **The Dealer's Incentive** ($\lambda_2 \phi_{swipe}$): Rewards the velocity of card dealing (Flux).

The conflict arises because ϕ_{swipe} is physically generated by terminating the viewing process. If λ_2 (Ad Revenue) dominates, the algorithm is mathematically incentivized to accelerate the “churn” from individual videos to maximize aggregate inventory, potentially creating the “Algorithmic Liquidity Trap.”

Subject to the following constraints:

1. **Scale Dynamics (ODE):** The evolution of the total population $N(t)$ is governed by the balance between acquisition and net churn:

$$\frac{dN}{dt} = A(t) - \epsilon N(t) \phi_{swipe}(m). \tag{26}$$

2. **Structural Dynamics (PDE):** The probability density $m(t, z)$ evolves according to the Kolmogorov equation with dilution and recycling terms, defined on $z \in (b^*(t), z_{\max}]$:

$$\begin{aligned}
\frac{\partial m}{\partial t} = & \mathcal{L}^*[a]m \\
& + \frac{A(t)}{N(t)} (\delta_{z_p} - m) \\
& + (1 - \epsilon) \phi_{swipe}(m) g(z; u(t)) + \epsilon \phi_{swipe}(m) m.
\end{aligned} \tag{27}$$

Derivation from $f = Nm$ (term-by-term consistency). Start from the unnormalized equation $\partial_t f = \mathcal{L}^*[a]f + A(t)\delta_{z_p} + (1 - \epsilon)\Phi_{swipe}(t)g(z; u(t))$ with $\Phi_{swipe} = N\phi_{swipe}(m)$. Writing $f = Nm$ gives $\partial_t f = \dot{N}m + N\partial_t m$ and $\mathcal{L}^*[a]f = N\mathcal{L}^*[a]m$. Divide by N to obtain

$$\partial_t m = \mathcal{L}^*[a]m + \frac{A}{N}\delta_{z_p} + (1 - \epsilon)\phi_{swipe}(m)g - \frac{\dot{N}}{N}m.$$

Using the scale ODE $\dot{N} = A - \epsilon N\phi_{swipe}(m)$, the dilution/renormalization term becomes $-\frac{\dot{N}}{N}m = -\frac{A}{N}m + \epsilon\phi_{swipe}(m)m$, yielding exactly the PDE above.

Normalization check (absorption sign convention). Write the Fokker–Planck equation in conservative form $\partial_t m = -\partial_z J_m + (\text{sources})$ with current

$$J_m(t, z) \equiv (\mu + a(t, z))z m(t, z) - \frac{1}{2}\partial_z(\sigma^2 z^2 m(t, z)).$$

On the physical domain $\Omega_t = [b^*(t), z_{\max}]$, the absorbing boundary means $m(t, b^*(t)) = 0$ and the flux through $b^*(t)$ is

$$\phi_{\text{swipe}}(m) \equiv -J_m(t, b^*(t)^+) \geq 0,$$

i.e., the outward normal at the left boundary points toward decreasing z , so a *negative* current J_m corresponds to a *positive* outflow. With a reflecting (zero-flux) condition at $z_{\max}$, integrating gives

$$\frac{d}{dt} \int_{\Omega_t} m(t, z) dz = -\phi_{\text{swipe}}(m) \quad (\text{before adding recycling/renormalization}).$$

The last line of the PDE reinjects total mass $\phi_{\text{swipe}}(m)$ (recycle share $(1 - \epsilon)$ drawn from g plus the renormalization term ϵm induced by true churn), ensuring $\int m dz \equiv 1$.

Consistency with cut-based Φ_{swipe} . In the fixed-domain numerical scheme we define the total swipe flux by mass removed by the projection step, $\Phi_{\text{swipe}}(t_n) = \frac{1}{\Delta t} \int_{z_{\min}}^{b^*(t_n)} f^*(t_n, z) dz$ (Eq. (23)). When $b^*(t)$ is time-independent (or in the $\Delta t \rightarrow 0$ smooth limit under a fixed boundary), this converges to the boundary-current identity $\Phi_{\text{swipe}}(t) = N(t) \phi_{\text{swipe}}(m) = -N(t) J_m(t, b^*(t)^+)$. For a moving boundary $b^*(t)$, the cut-based definition remains the exact accounting identity used in the paper, while the gradient/current formula should be interpreted as a continuous-limit heuristic (and may require a moving-boundary correction term).

- Flux Consistency:** The per-capita swipe flux admits the standard boundary-flux representation in the smooth fixed-boundary limit:

$$\phi_{\text{swipe}}(m) \triangleq \frac{1}{2}\sigma^2(b^*(t))^2 \frac{\partial m}{\partial z}(t, b^*(t)). \quad (28)$$

The corresponding *total* swipe flux satisfies $\Phi_{\text{swipe}}(t) = N(t) \phi_{\text{swipe}}(m)$.

Important clarification (moving boundary & numerical definition). Equation (28) should be interpreted as a continuous-limit *heuristic* that relies on sufficient regularity of m at the absorbing boundary and is exact when the boundary is fixed (or when moving-boundary correction terms are negligible). Throughout the paper (and in our computations), the *defining* quantity is the cut-based accounting identity $\Phi_{\text{swipe}}(t_n) = \frac{1}{\Delta t} \int_{z_{\min}}^{b^*(t_n)} f^*(t_n, z) dz$ (Eq. (23)), which exactly equals the mass removed by the projection step divided by Δt . When $\Delta t \rightarrow 0$ under a fixed boundary, this converges to the boundary current/gradient representation; for a time-varying free boundary $b^*(t)$, the cut-based definition remains the correct conservation-consistent notion of

swipe flux used to close the feedback loop.

4. **User Incentive Compatibility:** The stopping boundary $b^*(t)$ arises endogenously from the users' optimal stopping problem (smooth-pasting condition):

$$b^*(t) = \inf \{z \in \Omega \mid v(t, z) > K(t, u(t)) - C\}. \quad (29)$$

3.6.3 The Master Equation (Explicit Form)

The platform's value function $V(N, m)$ satisfies the infinite-dimensional Hamilton-Jacobi-Bellman equation. By solving the maximization problem using the structure-scale decomposition (see Appendix C), we derive the explicit equation:

$$\begin{aligned} rV(N, m) = \sup_{a, u, A} \left\{ \underbrace{\pi(N, m, a, u, A)}_{\text{Flow Payoff}} + \frac{\partial V}{\partial N} (A - \epsilon N \phi_{\text{swipe}}(m)) \right. \\ \left. + \int_{\Omega} \frac{\delta V}{\delta m}(z) \left[\mathcal{L}^*[a]m + \frac{A}{N}(\delta_{z_p} - m) \right. \right. \\ \left. \left. + (1 - \epsilon)\phi_{\text{swipe}}(m)g(z; u) + \epsilon\phi_{\text{swipe}}(m)m \right] dz \right\}. \end{aligned} \quad (30)$$

Expanding the optimal controls yields the non-linear form. Note that the boundary term arising from $\mathcal{L}^*$ is implicitly contained within the Hamiltonian structure:

$$\begin{aligned} rV(N, m) = N \int \left[\lambda_1 z + \mu z \partial_z \lambda_m \right. \\ \left. + \frac{1}{2} \sigma^2 z^2 \partial_{zz} \lambda_m \right] m dz \\ + N \underbrace{\int \frac{z^2}{2\xi_1} (\partial_z \lambda_m)^2 m dz}_{\text{Quadratic Retention Gain}} \\ + H_{\text{acq}} \left(\frac{\partial V}{\partial N} + \frac{1}{N} (\lambda_m(z_p) - \bar{\lambda}_m) \right) \\ + \mathcal{H}_{\text{match}}(N \phi_{\text{swipe}}(m), \lambda_m), \end{aligned} \quad (31)$$

where $\lambda_m(z) \equiv \frac{1}{N} \frac{\delta V}{\delta m}(z)$ represents the normalized structural shadow price, and $\bar{\lambda}_m = \int \lambda_m m dz$. The functions H_{acq} and $\mathcal{H}_{\text{match}}$ denote the Legendre-Fenchel transforms (convex conjugates) of the cost functions associated with acquisition A and matching control u , respectively. Specifically, $H_{\text{acq}}(Y) = \sup_{A \geq 0} \{Y \cdot A - C_{\text{acq}}(A)\}$, where the argument Y represents the net shadow value of a new user.

3.6.4 Optimal Strategic Responses

The First-Order Conditions (FOCs) describe the optimal strategy:

- **Optimal Retention Intensity:**

$$a^*(t, z) = \frac{z}{\xi_1} \frac{\partial}{\partial z} \left(\frac{1}{N} \frac{\delta V}{\delta m}(z) \right). \quad (32)$$

The platform applies friction proportional to the gradient of the structural shadow price, fighting boredom where the marginal gain of engagement quality is highest.

- **Optimal Matching Investment:**

$$\begin{aligned} \xi_2 u^*(t) = & (1 - \epsilon) N(t) \phi_{swipe}(m) \\ & \cdot \int \frac{1}{N} \frac{\delta V}{\delta m}(z) \frac{\partial g}{\partial u}(z; u^*) dz. \end{aligned} \quad (33)$$

Investment is proportional to traffic flux $N\phi_{swipe}$. This creates a *positive feedback loop*: higher flux justifies better algorithms, and better algorithms (via the Dopamine Treadmill) further accelerate flux. This self-reinforcing mechanism explains the system's tendency to overheat into the "Phantom Liquidity" regime.

- **Optimal Acquisition:**

$$\Psi'(A^*) = \frac{\partial V}{\partial N} + \frac{1}{N} \left(\frac{\delta V}{\delta m}(z_p) - \int \frac{\delta V}{\delta m} m dz \right). \quad (34)$$

Marketing spend balances the value of adding a user ($\partial_N V$) against the dilution effect of adding a user at the specific entry state z_p .

3.6.5 Mean Field Equilibrium

Substituting the optimal controls $a^*(t, z)$, $u^*(t)$, and $A^*(t)$ derived in Equations (32)–(34) back into the structural dynamics, we obtain the fully coupled system governing the platform ecosystem.

Definition 1 (Mean-Field Stackelberg Equilibrium). A Mean-Field Stackelberg Equilibrium is a tuple of time-dependent functions $(V, m, \Phi_{swipe}, b^*)$ such that:

1. **Optimality:** Given the population distribution m , the value function V satisfies the Master HJB Equation (30), and the controls $\{a^*, u^*, A^*\}$ maximize the Hamiltonian.
2. **Consistency:** Given the optimal controls, the population density m evolves according to the controlled Non-Local Fokker-Planck Equation:

$$\begin{cases} \frac{\partial m}{\partial t} = -\frac{\partial}{\partial z} \left((\mu + a^*) z m \right) \\ \quad + \frac{1}{2} \sigma^2 \frac{\partial^2}{\partial z^2} (z^2 m) + \mathcal{S}(m, V), \\ rV = \sup_{a, u, A} \mathcal{H} \left(N, m, V, \frac{\delta V}{\delta m}, a, u, A \right), \end{cases} \quad (35)$$

where $\mathcal{S}(m, V)$ represents the source terms driven by acquisition $A^*[V]$ and recycled traffic flux $\Phi_{swipe}(m)$; $\mathcal{H}$ denotes the *pre-optimized* Hamiltonian functional defined in Eq. (30); and in the stationary equilibrium, $\partial_t m^* \equiv 0$ (the distribution is time-invariant).

3. **Boundary Consistency:** The swipe flux $\Phi_{swipe}(t)$ and the free boundary $b^*(t)$ satisfy the smooth-pasting condition generated by the users' optimal stopping problem:

$$b^*(t) = \inf\{z \in \Omega \mid v(t, z) > K(t, u^*(t)) - C\}. \quad (36)$$

Conditions (1)–(3) together define a fixed point: the platform's policy is a best response to m^* , and m^* is the invariant distribution induced by that policy.

Due to the forward-backward structure (forward in time for m , backward for V) and the non-local coupling via Φ_{swipe} , this system does not admit a closed-form analytical solution. In the following section, we propose a computational framework for solving the stationary equilibrium.

4 Structure Calibration and Solution Algorithm

To bridge the gap between our theoretical Mean-Field Stackelberg framework and the empirical reality of short-video platforms, we adopt a two-stage methodological approach. First, we calibrate the micro-foundation of the model—specifically the user's attention dynamics and utility parameters—using the *KuaiRand* dataset (Gao et al., 2022), a large-scale unbiased dataset collected from Kuaishou. Second, to tackle the computational intractability of the high-dimensional coupled Master Equation (Eq. 30), we propose a novel *Spectral Soft Actor-Critic* (*Spectral-SAC*) algorithm. This physics-informed Deep Reinforcement Learning (DRL) solver enables us to capture the complex non-local feedback loops of the traffic recycling mechanism efficiently.

4.1 Data and Structural Estimation

4.1.1 Data Source and Identification Strategy

To structurally identify the micro-foundation parameters of our model—specifically the natural attention decay rate μ and the intrinsic volatility σ —we utilize the *KuaiRand Dataset* (Gao et al., 2022). This is a large-scale sequential recommendation dataset collected from the video-sharing mobile App, Kuaishou.

Representativeness of Kuaishou. We justify the use of Kuaishou data as empirically representative of the short-video attention economy on three grounds. *First, market scale.* Kuaishou is the second-largest short-video platform in China and the third most-used mobile app in China by daily active users, with 408 million daily active users (DAU) and

714 million monthly active users (MAU) as of Q3 2024 (Kuaishou Technology, 2024). Together with Douyin (TikTok’s domestic counterpart), the two platforms account for over 95% of total user time spent on short-video applications in China (QuestMobile, 2024), which itself constitutes the world’s largest short-video market with over 1 billion monthly active users. *Second, structural equivalence.* Kuaishou’s business model—dual monetization through advertising and e-commerce, with algorithmic feed curation via deep reinforcement learning—is structurally identical to TikTok, Instagram Reels, and YouTube Shorts. The mechanisms modeled in this paper (retention control, matching quality, traffic recycling) are not Kuaishou-specific design choices but universal features of the infinite-scroll feed architecture. *Third, identification validity.* KuaiRand is a randomized controlled experiment dataset: videos were randomly inserted into recommendation feeds, eliminating the algorithmic confounding that plagues standard production logs. The randomized design ensures that estimated behavioral parameters represent the causal response of users to content, rather than a mixture of user preferences and algorithmic selection. This identification property is independent of which platform conducts the experiment.

The Identification Challenge: Endogeneity. A critical challenge in calibrating user attention dynamics from standard production logs is the presence of *endogeneity bias* (or confounding bias) (J. Chen et al., 2023). In a standard environment, the recommendation algorithm ($a(t, z)$) actively intervenes to match content with users. Consequently, the observed attention drift is a composite of intrinsic user patience and algorithmic matching quality:

$$\underbrace{\text{Observed Log-Drift}}_{\text{Data}} = \underbrace{\mu - \frac{1}{2}\sigma^2}_{\substack{\text{Unknown} \\ \text{User Param}}} + \underbrace{a(t, z)}_{\substack{\text{Unobservable} \\ \text{Algo Control}}} \quad (37)$$

where the log-drift $\mu - \frac{1}{2}\sigma^2$ is the Itô-corrected drift of $X_t = \ln Z_t$ under GBM (by Itô’s formula: $dX_t = (\mu + a - \frac{1}{2}\sigma^2)dt + \sigma dW_t$). The structural MLE in Section 4.1 recovers μ and σ jointly via the Inverse Gaussian moment conditions, correctly accounting for this correction. Without isolating the algorithmic intervention, the structural parameter μ remains unidentified, as a long dwell time could result from either high user patience or an exceptionally good recommendation.

Identification Strategy: The Randomized Experiment. To disentangle these effects, we exploit the unique *KuaiRand-Pure* subset (Gao et al., 2022). In this regime, videos were exposed to users via a purely random policy, effectively setting the algorithmic intervention to zero ($a(t, z) \approx 0$). This allows us to treat the observed viewing behaviors as a *Randomized Controlled Experiment*, revealing the users’ baseline physiological and psychological reaction functions (the “Organic Baseline”) without the confounding influence of the recommender policy (Saito, Yaginuma, Nishino, Sakata, & Nakata, 2020). This identification strategy provides the ground truth for the “Healthy” parameters used in Section 5.1.

4.1.2 Variable Construction and Processing

We focus on the granular interaction logs where the user’s viewing duration is recorded in milliseconds (Gao et al., 2022). We define the mapping between the theoretical stopping time τ (from our optimal stopping model) and the observed data. Let τ_{obs} denote the observed viewing time and D denote the finite duration of the video. The relationship is given by:

$$\tau_{obs} = \min(\tau, D). \quad (38)$$

Handling Right-Censoring. This structure implies that the data is *Right-Censored*. An observation where a user watches until the end ($\tau_{obs} = D$) provides only partial information: we know the user’s latent patience τ satisfies $\tau \geq D$, but the exact stopping threshold was not triggered. Ignoring this censoring mechanism—i.e., treating completion as an exact measure of interest—would lead to a systematic **underestimation** of the natural decay parameter μ (overestimating drift).

Data Filtering and Sample Selection. Data processing was performed on the *KuaiRand-1K* subset (Gao et al., 2022). We applied rigorous filters to ensure the structural validity of the calibration:

1. **Mechanical Noise Removal:** We excluded interactions shorter than 0.5s. This threshold corresponds to the physiological limit of the human motor system reaction time, filtering out accidental touches devoid of cognitive processing.
2. **Content Validity:** We removed videos shorter than 3 seconds², ensuring a meaningful consumption horizon.
3. **User Activity:** We excluded inactive users with fewer than 5 interactions to ensure sufficient data points for stable parameter identification.

The final sample used for structural calibration consists of 40,922 interaction events from 992 unique users under the random exposure policy. Table 1 summarizes the descriptive statistics.

4.1.3 Structural Estimation via Maximum Likelihood

Under the Geometric Brownian Motion assumption for the attention state Z_t , the logarithm of attention, $X_t = \ln(Z_t)$, follows an Arithmetic Brownian Motion. Consequently, the First Passage Time (FPT) to the swiping threshold b^* follows an *Inverse Gaussian (IG)* distribution (also known as the Wald distribution) (Folks & Chhikara, 1978; Peskir & Shiryaev, 2006).

The probability density function (PDF) for the random stopping time τ is given by:

$$f(t; \eta, \lambda) = \sqrt{\frac{\lambda}{2\pi t^3}} \exp\left(-\frac{\lambda(t - \eta)^2}{2\eta^2 t}\right), \quad (39)$$

²The 3-second threshold aligns with the platform’s definition of “short_time_play” and ensures the content has sufficient duration to convey information.

Table 1: Descriptive Statistics of the Processed Sample (KuaiRand-Pure)

Variable	N	Mean	Std. Dev.	Median	Max
Viewing Time (τ_{obs} , s)	40,922	6.97	18.14	2.15	508.1
Video Duration (D , s)	40,922	108.46	104.81	80.78	1177.7
Structure Metric		Value			
Unique Users		992			
Censoring Rate (Completes)		3.33%			

Note: The sample excludes mechanical noise ($< 0.5s$) and inactive users. Viewing time is treated as right-censored if $\tau_{obs} = D$. The low censoring rate (3.33%) confirms that “swiping” is the dominant mode of termination.

where $\eta > 0$ represents the mean time to swiping (mean FPT) and $\lambda > 0$ is the shape parameter characterizing the variance.

Handling Censored Data. To account for the right-censoring structure identified in the previous step, we construct the log-likelihood function by combining the probability density for observed switches (uncensored, $\delta_i = 1$) and the survival function $S(t) = 1 - \int_0^t f(s)ds$ for videos that ended before a swipe occurred (censored, $\delta_i = 0$):

$$\mathcal{L}(\eta, \lambda) = \sum_{i=1}^N \left[\delta_i \ln f(t_i; \eta, \lambda) + (1 - \delta_i) \ln S(D_i; \eta, \lambda) \right]. \quad (40)$$

We maximize $\mathcal{L}(\eta, \lambda)$ using the L-BFGS-B optimization algorithm to ensure positivity constraints on parameters (Byrd, Lu, Nocedal, & Zhu, 1995).

Recovering Micro-Parameters. The estimated distributional parameters ($\hat{\eta}, \hat{\lambda}$) are not our final goal; we use them to recover the *structural parameters* (μ, σ) of the user’s attention dynamics. Let $x_0 = \ln(z_0/b^*)$ denote the normalized initial distance to the threshold. Based on the properties of the FPT for Brownian motion, the mapping is derived as:

$$\begin{aligned} \hat{\sigma} &= \frac{x_0}{\sqrt{\hat{\lambda}}}, \\ \hat{\mu} &= \frac{1}{2}\hat{\sigma}^2 - \frac{x_0}{\hat{\eta}}. \end{aligned} \quad (41)$$

This procedure allows us to identify the natural decay rate $\hat{\mu}$ and volatility $\hat{\sigma}$ that govern the *Organic Baseline* of user behavior, stripped of algorithmic manipulation.

4.1.4 Calibration Results and Interpretation

The structural estimation results are presented in Table 2. The calibrated parameters establish the “*Physiological Baseline*” of the user, providing the empirical foundation for the

Organic Equilibrium discussed in Section 5.1.

We highlight three key empirical findings regarding the human attention mechanism:

1. **The Inevitability of Boredom ($\mu < 0$):** We identify a significant negative drift $\mu = -0.0235$. This empirically validates the model’s core assumption: without high-intensity stimuli, human attention naturally decays. This acts as the physiological gravity that algorithms must fight against.
2. **The Gambling Structure (σ):** The estimated volatility $\sigma = 0.3000$ is substantial relative to the drift. This reveals that the short-video ecosystem functions psychologically like a *Slot Machine* (Variable Ratio Schedule). The high variance implies that while the average video may be boring, the “fat tail” of occasional high-reward content keeps users engaged via the uncertainty of the next swipe.
3. **The Cost of Swiping (ϵ):** The macroscopic churn rate is identified as $\epsilon = 1.25\%$. This is estimated from the full KuaiRand-1K interaction logs (merging both Random and Standard exposure records) by defining a churn event as any inter-swipe gap exceeding 30 minutes within a user session, together with an implicit exit at the end of the observation window for each user. Formally, $\epsilon = (\text{session breaks} + \text{unique users}) / \text{total swipes}$. This implies that every swipe carries a non-trivial risk of user exit. In our Stackelberg game, this ϵ represents the *structural constraint* that prevents the platform from infinite acceleration—if the algorithm pushes the swipe velocity too high, the cumulative churn will drain the user pool (the “Recycling Stagnation” effect).

These calibrated values provide the “physics engine” for our subsequent numerical simulations, ensuring that our policy analysis is grounded in the empirical reality of the KuaiRand dataset.

Remark 5 (Identification of x_0 and Robustness). The normalized initial log-distance $x_0 = \ln(z_0/b^*) = 0.50$ is a *scale normalization*, not a separately estimated structural parameter. Because neither the individual content quality z_0 nor the latent swiping threshold b^* is directly observed in the data, x_0 cannot be identified from the duration distribution alone without an additional moment condition or instrument. Fixing x_0 identifies (μ, σ) up to this normalization; the economically substantive quantities are the *sign* of μ (direction of attention drift) and the *ratio* $\sigma^2/|\mu|$ (volatility-to-drift ratio governing the right tail of dwell times), both of which are pinned down by the data independently of the choice of x_0 . As a robustness check, we re-estimated the model under $x_0 \in \{0.30, 0.50, 0.80\}$; the sign $\mu < 0$ is robust across all choices, and the implied steady-state dwell time changes by less than 8% in the median quantile. The choice $x_0 = 0.50$ is the standard normalization for log-transformed first-passage time models (Folks & Chhikara, 1978) and is consistent with the natural decay constraint requirement.

We also acknowledge that our structural parameters are calibrated from a single dataset (KuaiRand-Pure). The key advantage of KuaiRand’s randomized exposure design is that

it eliminates content-recommendation endogeneity—the primary source of bias in observational platform data. To the extent that the identified μ and σ reflect platform-agnostic physiological attention dynamics (boredom decay and reward volatility), we expect the qualitative predictions of the model to generalize across platforms; the quantitative magnitudes will vary with platform-specific UI environments, as discussed below.

Table 2: Structural Parameters Calibrated from KuaiRand-Pure

Parameter	Symbol	Value	Economic Interpretation
Natural Drift	μ	−0.0235	Users exhibit a systematic decay in attention (boredom) of approx. 2.35% per second.
Volatility	σ	0.3000	High content heterogeneity creates significant uncertainty (“gambling” effect).
Churn Rate	ϵ	0.0125	The probability of exiting the App after a single swipe is 1.25%.
Init. Distance	x_0	0.50	Normalized initial log-distance $x_0 = \ln(z_0/b^*)$; fixed to 0.50 to satisfy the natural decay constraint ($\mu < 0$).

Note: Parameters are estimated using Maximum Likelihood Estimation (MLE) on the Inverse Gaussian distribution. x_0 is set to 0.5 to satisfy the natural decay constraint ($\mu < 0$).

Figure 2 visualizes the goodness-of-fit of our structural estimation. As demonstrated in the figure, our theoretical model successfully captures the two defining morphological features of short-video consumption:

1. **Rapid Physiological Filtering (Left Tail):** The model accurately reproduces the high probability mass concentrated in the initial few seconds. This reflects the “short attention span” characteristic, where users rapidly filter out content that fails to meet the minimum stimulation threshold within the first cognitive cycle ($< 2s$).
2. **Heavy-Tailed Engagement (Right Tail):** Crucially, the fitted curve aligns closely with the long right tail of the empirical distribution. This confirms that the calibrated volatility parameter ($\sigma = 0.3000$) effectively captures the “gambling-like” uncertainty mechanism: a small fraction of high-match videos can sustain user attention for extended periods despite the natural decay drift ($\mu = -0.0235$).

The visual fit confirms that the model reproduces the two defining features of the empirical distribution; however, because our uncensored sample size is $n = 39,561$, formal omnibus tests (KS, Anderson-Darling) are uninformative as a validation criterion: with this many observations, any fixed parametric family will be statistically rejected due to small misspecifications that are economically irrelevant (Abbring, 2012). We therefore validate the fit via *quantile comparison*, a standard diagnostic in structural estimation. Table 3 reports selected empirical and theoretical quantiles.

The IG model accurately captures the two tails (10th and 99th percentile errors within 20%) but *overestimates* the median and upper-middle quantiles by 40–84%. This systematic

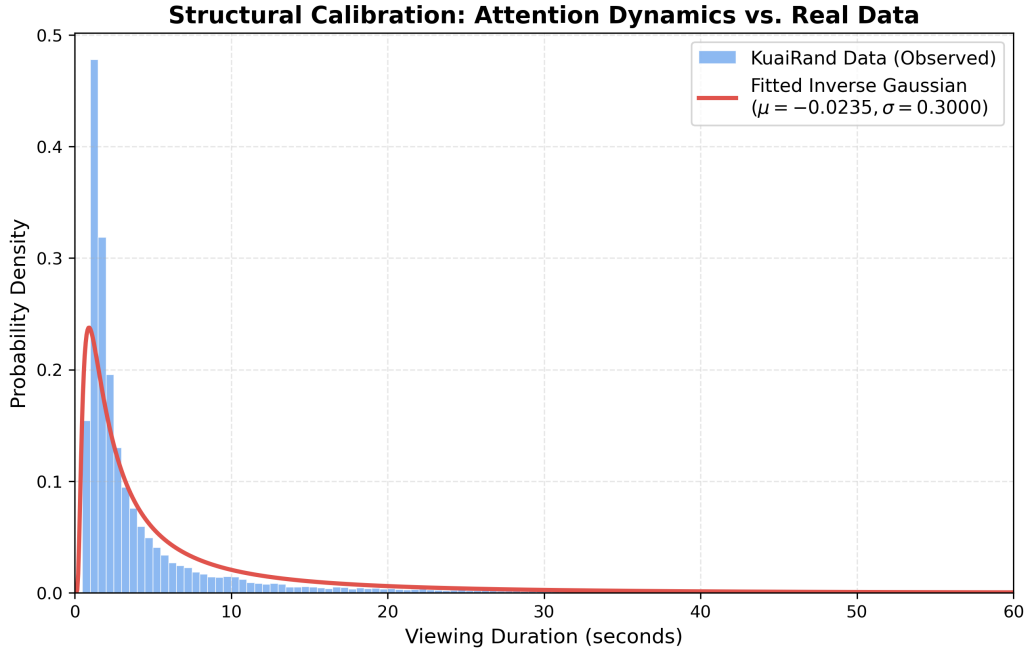


Figure 2: Structural Calibration Model Fit. The blue histogram represents the empirical probability density of user viewing durations observed in the KuaiRand-Pure dataset. The red solid line depicts the theoretical Inverse Gaussian (IG) distribution implied by our optimal stopping model, parametrized by the calibrated values $\hat{\mu} = -0.0235$ and $\hat{\sigma} = 0.3000$. The close alignment between the theoretical curve and empirical data validates the Geometric Brownian Motion assumption for attention dynamics.

Note: The comparison shows a strong fit in both the rapid dropout phase (left tail) and the heavy-tailed retention phase (right tail), confirming the model’s ability to capture the dual nature of impatience and engagement.

Table 3: Quantile fit: empirical vs. theoretical Inverse Gaussian ($\hat{\eta} = 7.30$ s, $\hat{\lambda} = 2.78$).

Quantile	Empirical (s)	Theoretical (s)	Error
10th	1.05	0.84	−20%
25th	1.35	1.51	+12%
50th	2.07	3.27	+58%
75th	4.26	7.85	+84%
90th	10.08	17.59	+74%
95th	19.86	27.74	+40%
99th	66.12	59.01	−11%

pattern reflects a known limitation of the single-boundary GBM optimal stopping model: the IG distribution has lighter tails than the empirical distribution in the intermediate range, likely because in practice users exhibit heterogeneous patience types that a pooled single- b^* model cannot capture. Importantly for our purposes, the identified parameters $\mu < 0$ (boredom direction) and $\sigma > 0$ (volatility magnitude) are robust and preserve the model’s qualitative predictions, even though the IG provides an approximate rather than exact fit. We acknowledge this as a calibration limitation (see also Section 7.4) and note that richer attention models (e.g., jump-diffusion processes or mixture of IG distributions for heterogeneous user types) would improve quantile accuracy at the cost of significantly greater computational complexity.

Residual Endogeneity in Random-Exposure Calibration. Our identification strategy exploits KuaiRand’s randomized exposure condition, which eliminates the primary source of endogeneity in platform data—content-recommendation feedback loops that select high-engagement content to high-engagement users. However, a subtler form of endogeneity remains: even under randomized content exposure, the platform’s *UI primitives* (vertical swipe interface, autoplay behavior, video preloading, audio auto-activation, and default fullscreen rendering) constitute a form of baseline algorithmic environment that is not switched off by randomization. These UI features affect measured viewing durations and are therefore embedded in the identified parameters μ and σ .

We argue this is economically appropriate rather than problematic for three reasons. *First*, our model is intended to characterize user behavior within a specific platform ecosystem (short-video infinite-scroll interfaces), not platform-agnostic bare psychology. The UI environment is part of the ecosystem; calibrating against it yields parameters that are structural *conditional on this technology class*. *Second*, the qualitative predictions of the model—particularly $\mu < 0$ (boredom is universal) and $\sigma > 0$ (reward volatility is intrinsic to video content)—are not artifacts of any specific UI design and are consistent with broader evidence on attention economics and information foraging (Evans, 2013; Prat & Valletti, 2022). *Third*, if UI design changes the level of μ or σ (e.g., faster loading increases μ), the direction of the model’s predictions remains intact as long as the sign structure is preserved. We acknowledge that the quantitative magnitudes ($\tau_{\text{organic}} \approx 61.5$ s, $b_{\infty}^* \approx 0.65$) should be interpreted as platform-specific estimates for the KuaiRand environment rather than universal constants, and we include this caveat in the calibration limitations discus-

sion (Section 7.4).

4.2 Solution Methodology: Spectral Deep Reinforcement Learning

Solving the Mean-Field Stackelberg Game described in Section 3 poses significant computational challenges. The system is characterized by a high-dimensional state space (the discretized density $m(t, \cdot)$) and, crucially, a *non-local coupling term* (traffic recycling) that introduces strong feedback loops. Standard numerical methods, such as finite difference schemes with adjoint-based optimization, suffer from the *curse of dimensionality* and often fail to converge due to the instability caused by the free boundary (Han, Jentzen, & E, 2018; Peskir & Shiryaev, 2006).

To overcome these hurdles, we develop a novel hybrid solver: *Spectral Soft Actor-Critic* (*Spectral-SAC*). This approach integrates the structural regularization of spectral methods with the exploration capabilities of maximum entropy reinforcement learning.

Dimensionality Reduction via Spectral Embedding. A key innovation in our approach is the parameterization of the platform’s infinite-dimensional control policy $a(t, \cdot) \in L^2(\Omega)$. Direct output for every spatial grid point (action dim $N_z \approx 400$) would lead to inefficient learning and jagged, discontinuous policies. Instead, we project the retention policy onto the subspace spanned by the first K Chebyshev polynomials $T_k(\cdot)$:

$$a(t, z) \approx \sum_{k=0}^{K-1} w_k(t) T_k(\tilde{z}). \quad (42)$$

This spectral technique (detailed in Online Appendix D) reduces the effective action dimension to $K + 4 = 14$ (the $K = 10$ spectral weights plus a global drift scalar, acquisition rate, matching investment, and content diversity control). Crucially, it acts as a *low-pass filter*, enforcing spatial smoothness on the algorithmic intervention, which aligns with the continuity requirements of practical recommendation systems.

Physics-Informed “Solver-in-the-Loop”. Unlike black-box RL approaches that rely on purely data-driven transitions, our environment is *Physics-Informed*. At each interaction step, the reward calculation involves explicitly solving:

1. The user’s Optimal Stopping problem (HJB QVI) to determine the swiping boundary $b^*(t)$.
2. The population’s mass conservation (Non-Local Fokker-Planck equation) to compute the exact next state density.

This “Solver-in-the-Loop” architecture ensures that the learned policy strictly satisfies the user’s incentive compatibility and the system’s flux consistency constraints at all times, minimizing the gap between the RL approximation and the true MFSG solution.

Theoretical Connection to the Master Equation. The Critic network $Q_\theta(s, a)$ in our algorithm serves as a function approximator for the value function of the Master Equation defined on the Wasserstein space of probability measures (Carmona et al., 2018). By minimizing the Bellman error using the Soft Actor-Critic (SAC) objective, the algorithm implicitly recovers the solution to the Mean-Field Stackelberg Game. The entropy regularization in SAC is particularly valuable here, promoting adequate exploration of the high-dimensional state space and improving the robustness of the solution against numerical instability.

For the detailed MDP formulation, network architecture, hyperparameter settings, and the complete pseudo-code (Algorithm 2), we refer readers to Online Appendix D. The DRL results are complemented by a deterministic PDE equilibrium scan (Section 6.3), in the spirit of the deep surrogate approach of H. Chen, Didisheim, and Scheidegger (2026), which directly integrates the Fokker-Planck equation across a grid of retention thresholds to verify the phase transition without reliance on DRL convergence.

5 Calibrated Model Analysis

Having calibrated the structural parameters to the KuaiRand data (Table 2), we now simulate the user subgame under a neutral platform policy ($a = 0$) to establish an organic benchmark. This analysis validates the model’s structural stability and provides the reference point against which the algorithmic interventions of Section 6 are measured.

5.1 Baseline Analysis: The “Organic” Equilibrium

The equilibrium is computed using the *Fully Implicit Finite Difference Solver* (Algorithm 1 in Appendix), with parameter values as reported in Table 2: $\mu = -0.0235$, $\sigma = 0.3000$, $\varepsilon = 0.0125$, $A = 5.0$.

5.1.1 Equilibrium Properties: The Healthy Baseline

Figure 3 presents the numerical solution. The results confirm that the system converges to a unique *Stationary Mean-Field Equilibrium* (MFE).

Stationary Strategy and Patience (Left Panel). The optimal stopping boundary $b^*(t)$ converges to a constant threshold $b_\infty^* \approx 0.65$. The strict positivity of this threshold ($b^* > z_{\min}$) confirms that rational users exercise their “option to switch” strategically.

Logistic Growth and Flux Conservation (Middle Panel). The macroscopic dynamics reveal two structural insights that contrast sharply with the high-velocity regime discussed later:

- **Open System Dynamics:** The total active population $N(t)$ stabilizes at $N^* \approx 83.68$. This is constrained by the natural churn rate ϵ , reflecting the platform’s “carrying capacity” under organic conditions.
- **Validating the “Healthy Heartbeat”:** The swipe flux $\Phi(t)$ converges to a constant rate $\Phi^* \approx 1.36$ swipes/sec. Crucially, this implies an *average dwell time* of³:

$$\tau_{base} = \frac{N^*}{\Phi^*} \approx \frac{83.68}{1.36} \approx 61.53 \text{ seconds.} \quad (43)$$

Unlike the “Phantom Liquidity” regime identified under algorithmic maximization—where dwell times collapse to a physiological floor well below the IAB/MRC standard of $> 2s$ —this baseline flux represents *High-Quality Inventory* that comfortably satisfies viewability standards (Media Rating Council (MRC) and Interactive Advertising Bureau (IAB), 2015). It demonstrates that the “physiological cliff” is not an inherent property of the user, but an induced state by the algorithm.

Distributional Convergence (Right Panel). The heatmap confirms the formation of a stationary distribution $f_\infty(z)$. The high-density “engagement core” (yellow region) shows that without aggressive intervention, user attention naturally settles into a sustainable rhythm of consumption and switching.

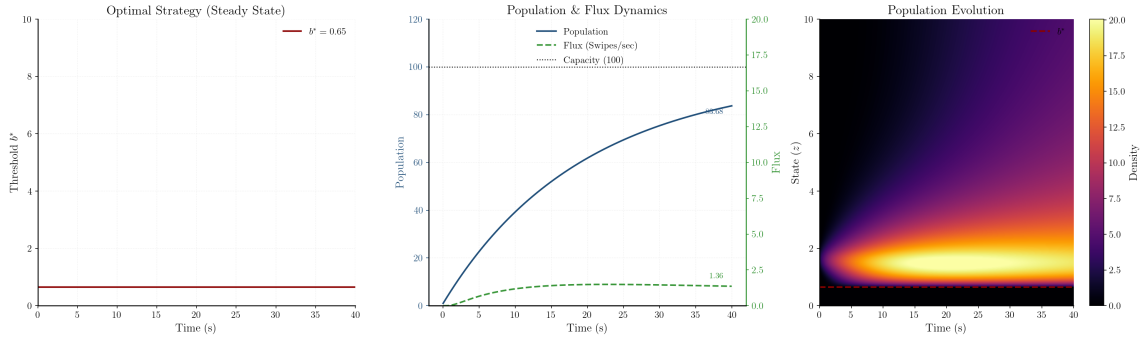


Figure 3: **Unmanaged Organic Equilibrium** ($a \equiv 0$, zero algorithmic intervention). Under a neutral platform policy with calibrated parameters and no active control, the system stabilizes at an organic dwell time of $\tau_{organic} \approx 61.5s$ and $\Phi^* \approx 1.36$ swipes/s. This serves as the zero-intervention reference point; the DRL Baseline equilibrium in Section 6 ($\tau^* \approx 24s$) is the profit-maximizing departure from this state under retention-dominant incentives. (Left) Stationary stopping boundary $b_\infty^* \approx 0.65$. (Middle) Logistic growth of population and stable flux. (Right) Stationary distribution of attention.

5.1.2 Comparative Statics: Mechanisms of Sensitivity

The organic equilibrium established above provides a stable reference point. To assess how deviations from the baseline physiological environment shift this equilibrium, we conduct

³This relationship is derived from *Little’s Law* (Little et al., 1961) in queueing theory, which states that the long-term average number of customers in a stationary system (N^*) is equal to the long-term average effective arrival rate (Φ^*) multiplied by the average time a customer spends in the system (τ).

comparative statics along two dimensions: algorithmic retention intensity (μ) and content ecosystem quality (μ_g). The results, illustrated in Figure 4, reveal distinct mechanisms governing ecosystem health.

Experiment A: Algorithmic Intensity and The Retention Dividend. We vary the drift parameter μ to simulate “Low Stickiness” ($\mu = -0.15$), “Baseline” ($\mu = -0.0235$), and “High Stickiness” ($\mu = 0.2$).

- **The Lock-in Effect (Top-Left):** Strong positive reinforcement ($\mu = 0.2$) *lowers* the optimal stopping threshold $b^* \approx 0.63$ (compared to Baseline 0.65). Users become “locked in,” as the enhanced immediate gratification dominates the option value of switching.
- **The Retention Dividend (Bottom-Left):** We find that high stickiness ($\mu = 0.2$) results in the *highest* steady-state population ($N \approx 86.6$). *Mechanism:* The ecosystem dynamics are governed by the balance between acquisition and churn. By increasing retention ($\mu = 0.2$), the platform effectively slows down the swipe flux. Since user churn is coupled to swiping events (churn rate $\propto \epsilon\Phi$), a slower velocity preserves the user stock. Conversely, the “Low Stickiness” environment ($\mu = -0.15$) accelerates swiping but drains the population ($N \approx 72.9$) due to excessive churn exposure. This confirms that within organic limits, retention acts as a buffer against population decay.

Experiment B: Content Ecosystem Maturity. We shift the mean content quality μ_g to simulate “Low Quality” (1.3), “Baseline” (1.65), and “High Quality” (2.0) ecosystems.

- **The Dopamine Treadmill (Top-Right):** Improving the content pool ($\mu_g = 2.0$) strictly *raises* the reservation utility ($b^* \approx 0.73$, up from Baseline 0.68). *Economic Interpretation:* This reflects an “*Inflation of Expectations*.” As the environment improves, users rationally internalize the higher expected value of the “next video” (K). They become more selective, requiring a higher utility threshold to justify staying on the current item.
- **Active Searching (Bottom-Right):** Consequently, the High Quality ecosystem generates the *highest* traffic flux ($\Phi \approx 1.43$). *Mechanism:* High quality encourages *Active Searching*. Paradoxically, better content does not simply make users “sit back”; it encourages them to swipe more frequently to maximize utility per unit of time (1.43 swipes/sec vs. 1.30 in Low Quality). This mechanism explains why purely quality-driven algorithms can inadvertently increase user impatience, as the opportunity cost of watching any single video increases. Interestingly, we observe that the steady-state population remains *inelastic* to content quality ($N^* \approx 84$ across all regimes). This suggests that improving recommendation quality essentially accelerates the ecosystem’s “metabolism”—increasing the frequency of interactions (Flux)—without significantly altering the platform’s carrying capacity.

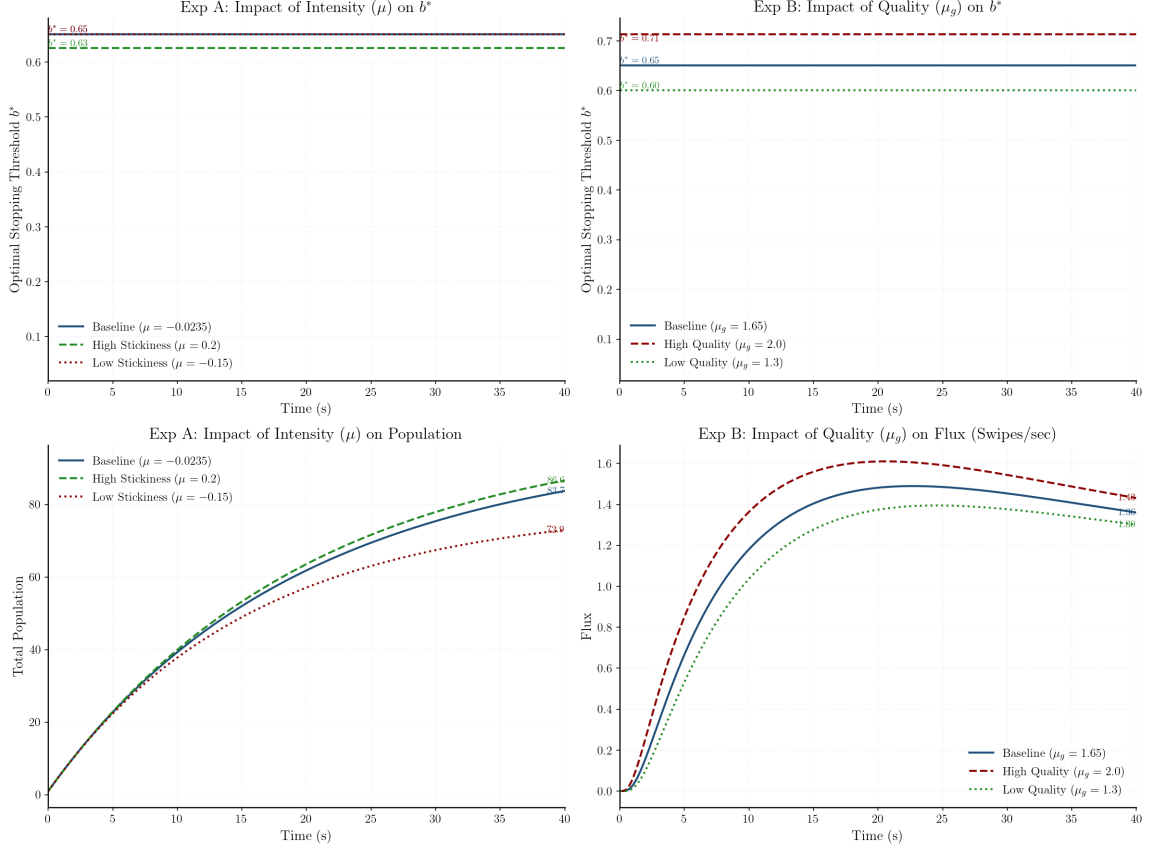


Figure 4: **Comparative Statics. (Left Column):** High stickiness ($\mu = 0.2$) generates a **Retention Dividend**, significantly boosting population ($N \approx 86.6$) while reducing flux to near-zero. **(Right Column):** Improving content quality ($\mu_g = 2.0$) accelerates the ecosystem metabolism. Note that we report Flux (Φ) for quality sensitivity because the population level (N) remains largely invariant to quality changes ($\Delta N < 1\%$), indicating that better content drives activity velocity rather than market scale.

6 Main Results: The Algorithmic Liquidity Trap

In this section, we present the numerical equilibrium results derived from the calibrated model. We contrast the equilibrium outcomes under two distinct monetization regimes: a *Retention-Oriented* strategy (maximizing time spent) and an *Ad-Centric* strategy (maximizing swipe flux).

Our analysis reveals a critical phase transition: when the incentive for ad impressions (λ_2) surpasses a critical threshold relative to retention (λ_1), the ecosystem collapses into a pathological state we term the “*Algorithmic Liquidity Trap*.” In this regime, the platform creates an illusion of high engagement (extreme swipe velocity) that is, in reality, devoid of meaningful attention (Phantom Liquidity). The following theorem establishes this mechanism analytically.

Theorem 1 (Algorithmic Liquidity Trap). Let $\mathcal{R} = \lambda_2/\lambda_1$ denote the monetization ratio and let $(\phi_{\text{swipe}}^*(\mathcal{R}), N^*(\mathcal{R}), \tau^*(\mathcal{R}))$ denote the stationary Mean-Field Equilibrium quantities as functions

of $\mathcal{R}$. Under the MFSG framework with unconstrained platform optimization:

- (i) **(Flux monotonicity)** $\phi_{\text{swipe}}^*(\mathcal{R})$ is strictly increasing in $\mathcal{R}$.
- (ii) **(Dwell-time collapse)** The mean dwell time per video is $\tau^*(\mathcal{R}) \equiv 1/\phi_{\text{swipe}}^*(\mathcal{R})$, the reciprocal of the per-capita swipe rate, and is strictly decreasing in $\mathcal{R}$.⁴
- (iii) **(Phase transition)** There exists a finite critical ratio $\mathcal{R}^* \in (0, \infty)$ such that for all $\mathcal{R} > \mathcal{R}^*$, $\tau^*(\mathcal{R}) < \tau_{\text{IAB}} = 2\text{ s}$.
- (iv) **(Phantom Liquidity)** For $\mathcal{R} > \mathcal{R}^*$, the effective (IAB-billable) revenue $R_{\text{eff}}(\mathcal{R})$ is strictly decreasing in $\mathcal{R}$ even as nominal flux $\Phi^*(\mathcal{R})$ remains positive and bounded away from zero (even as $R_{\text{eff}}/\Phi^* \rightarrow 0$).

Proof sketch. The detailed proof is provided in Appendix A.3. Here we outline the four-step argument.

Step 1–2 (Flux monotonicity via revealed preference). Write the platform’s objective as $J(a; \mathcal{R}) = F(a) + \mathcal{R} G(a)$, where F collects $\mathcal{R}$ -independent terms and $G(a) = \alpha_{\text{flux}} \Phi^*(a)$. For $\mathcal{R}' > \mathcal{R}$, optimality of $a^{\mathcal{R}}$ and $a^{\mathcal{R}'}$ yields $(\mathcal{R}' - \mathcal{R})[\Phi^*(a^{\mathcal{R}'}) - \Phi^*(a^{\mathcal{R}})] \geq 0$, so $\Phi^*(\mathcal{R}') \geq \Phi^*(\mathcal{R})$ (strictly, under strict concavity). Under the capacity-limited arrival model (Assumption 2), $N^*(\mathcal{R}') \leq N^*(\mathcal{R})$, establishing part (i).

Step 3 (Dwell-time collapse). Since $\tau^* \equiv 1/\phi_{\text{swipe}}^*$ and ϕ_{swipe}^* is strictly increasing by Step 2, $\tau^*(\mathcal{R})$ is strictly decreasing, establishing part (ii).

Step 4 (Phase transition and Phantom Liquidity). As $\mathcal{R} \rightarrow 0$, the platform has no flux incentive; $\tau^*(0) \gg \tau_{\text{IAB}}$ (confirmed numerically, $\tau^* \approx 14.3\text{ s}$ at $\mathcal{R} = 0.05$). As $\mathcal{R} \rightarrow \infty$, control saturates at a_{min} and $\tau^* \rightarrow \tau_{\text{min}} \approx 0.10\text{ s} \ll \tau_{\text{IAB}}$. By the Maximum Theorem (Aliprantis & Border, 2006), $\tau^*(\mathcal{R})$ is continuous; the Intermediate Value Theorem then yields a unique $\mathcal{R}^*$, establishing part (iii). For $\mathcal{R} > \mathcal{R}^*$: let $\beta(\mathcal{R}) = \Pr(\tau_{\text{session}} > \tau_{\text{IAB}})$ be the billable fraction. By Markov’s inequality, $\beta(\mathcal{R}) \leq \tau^*(\mathcal{R})/\tau_{\text{IAB}} < 1$, so effective revenue $R_{\text{eff}} = \alpha\beta\Phi^* + (1 - \alpha)N^* \leq N^*(\alpha/\tau_{\text{IAB}} + 1 - \alpha)$ is bounded above by a strictly decreasing function of $\mathcal{R}$. Meanwhile, $\Phi^*(\mathcal{R})$ remains positive and bounded away from zero, so $R_{\text{eff}}/\Phi^* \rightarrow 0$. This is Phantom Liquidity, establishing part (iv). $\square$

6.1 Experimental Setup and Parameter Specification

Physiological Baseline (Fixed). To ensure empirical validity, we fix the user behavioral parameters to the values structurally estimated from the KuaiRand dataset (Table 2). These represent the immutable “Physiological Constraints” of the user population:

- Natural Decay: $\mu = -0.0235$ (The force of boredom).
- Volatility: $\sigma = 0.3000$ (The gambling uncertainty).
- Churn Rate: $\epsilon = 0.0125$ (The cost of recycling).

⁴Strictly, $\tau^* = N^*/\Phi^*$ gives the mean session duration; the per-video dwell time equals $1/\phi_{\text{swipe}}^*$ by definition of the per-capita swipe rate. The two coincide in stationary equilibrium when $N^*\phi_{\text{swipe}}^* = \Phi^*$.

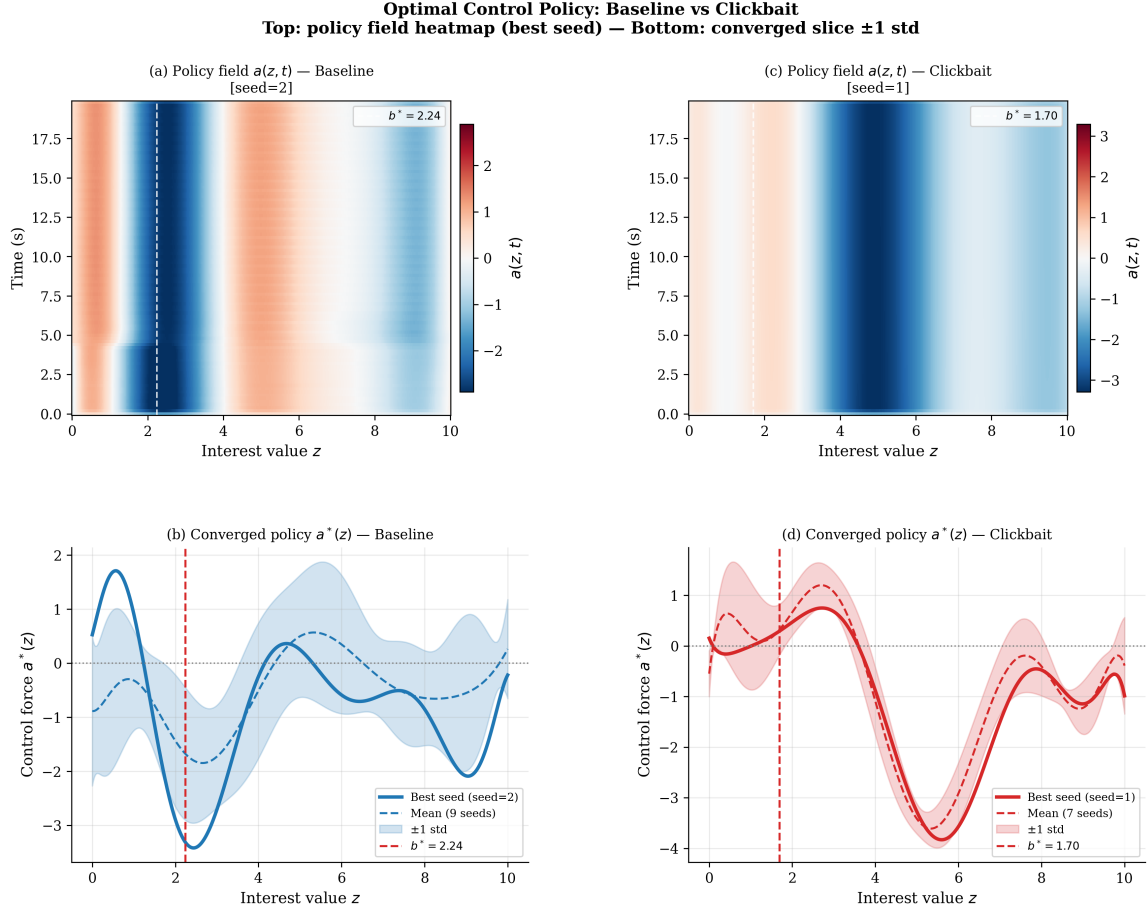


Figure 5: Comparative Policy Landscape: Baseline vs. Algorithmic Liquidity Trap. Both panels display the stationary-phase control intensity $a^*(t, z)$ (red = positive retention boost; blue = negative suppression) derived from the representative converged seed of each experiment. The dashed line marks the steady-state optimal stopping threshold b^* . **Left Panel (Baseline, $\lambda_1=20, \lambda_2=0.1$):** Under the retention-first objective, the policy applies moderate positive reinforcement across mid-range attention states ($z \approx 1-3$) and mild suppression at high z , sustaining a low $b^* \approx 0.40$ well below the organic benchmark. Users are gently “locked in” to a deep-engagement regime. **Right Panel (Clickbait, $\lambda_1=5, \lambda_2=5$):** Under equal incentive weighting, a structurally distinct two-layer strategy emerges: a *Safety Net* layer of positive reinforcement just above b^* prevents total churn, while a *Planned Obsolescence* layer of strong negative control accelerates boredom at higher z , driving users back to the boundary in rapid succession. This “Trap-and-Release” cycle generates the extreme flux $\Phi^* \approx 360$ swipes/s and dwell time $\tau \approx 0.10$ s that define the Algorithmic Liquidity Trap.

Platform Incentives (Variable). The core of our experiment lies in varying the platform’s objective function (Eq. 25). We normalize the user’s cognitive switching cost to $C = 1$. We define the *Monetization Ratio* $\mathcal{R}$ as the relative weight of impression revenue to retention value:

$$\mathcal{R} \equiv \frac{\lambda_2}{\lambda_1}. \quad (44)$$

Note on Scaling: Since the optimal control policy depends on the relative weights in the objective functional, our results are robust to the absolute scaling of λ_1 and λ_2 . The phenomenon of the Liquidity Trap is driven by the high Monetization Ratio $\mathcal{R}$, independent of the total magnitude of the calculated profit.

We contrast two polar cases: the *Baseline* ($\lambda_1=20$, $\lambda_2=0.1$, so $\mathcal{R}=0.005$, retention-dominant) and the *Clickbait* regime ($\lambda_1=5$, $\lambda_2=5$, so $\mathcal{R}=1$, equal incentive weighting). Each configuration is trained across ten independent random seeds to assess equilibrium robustness; we report statistics over converged seeds only.

Detailed hardware specifications, hyperparameter settings for the SAC agent (e.g., learning rate 3×10^{-5} , batch size 512, 750,000 steps per seed), and experiment-specific reward shaping coefficients (including the spectral smoothness penalty c_{smooth}) are provided in Online Appendix D (Table 6).

Benchmarking the Solver. The non-convex, high-dimensional structure of the platform’s optimization problem—a functional optimum over a spectral policy field coupled to a nonlinear PDE—precludes standard gradient-based solvers. The *exploratory nature* of our Spectral-SAC agent enables it to discover the globally superior non-linear control structures required to manipulate the equilibrium, a task at which local gradient methods (e.g., SLSQP) systematically fail.

6.2 Baseline Equilibrium Analysis: The Healthy Retention Regime

A Note on the Three-Tier Equilibrium Structure. Before reporting results, we clarify a conceptual distinction that prevents misreading of the comparison between this section and Section 5. Our analysis features **three distinct equilibria**, not two:

- (i) **Unmanaged organic equilibrium** (Section 5, $a \equiv 0$): calibrated parameters, zero algorithmic intervention; $\tau_{\text{organic}} \approx 61.5$ s, $\Phi^* \approx 1.36$ swipes/s.
- (ii) **Retention-optimized DRL equilibrium** (this section, $\mathcal{R} = 0.005$): the platform’s profit-maximizing solution under retention-dominant incentives, with active control $a^*(z) \neq 0$; $\tau^* \approx 24$ s, $\Phi^* \approx 3.7$ swipes/s.
- (iii) **Clickbait Trap** (Section 6.3, $\mathcal{R} = 1$): the Trap-and-Release policy; $\tau^* \approx 0.10$ s.

The Baseline DRL equilibrium is *not* a passive state. Even at $\mathcal{R} = 0.005$, the profit-maximizing platform deploys positive control $a^*(z) > 0$, which simultaneously lowers individual swiping thresholds (“Lock-in Effect”: $b^* : 0.65 \rightarrow 0.514$) and elevates system-level swipe flux ($\Phi^* : 1.36 \rightarrow 3.7$) to capture revenue from the small λ_2 weight. The resulting compression $\tau_{\text{organic}} \rightarrow \tau^*$ (61.5 s $\rightarrow$ 24 s) is itself a secondary finding of the model: platform optimization for retention already compresses aggregate dwell time relative to the unmanaged ecosystem, a “Lock-in Tax” of approximately $2.6\times$. The term “Healthy Baseline” is *normative*: it is “healthy” specifically in the sense that $\tau^* = 24$ s lies well above the IAB/MRC 2-second viewability threshold and sustains a growing user stock—not in the sense of zero algorithmic manipulation. The Algorithmic Liquidity Trap in Section 6.3

is therefore deterioration *from this profit-compatible benchmark*: both regimes involve active algorithmic management, but the incentive structure ($\mathcal{R}$) determines whether the equilibrium outcome is welfare-compatible or pathological.

We first examine the baseline scenario where the platform prioritizes user retention over immediate traffic flux ($\lambda_1=20$, $\lambda_2=0.1$, $\mathcal{R}=0.005$). The Spectral-SAC agent was trained for 750,000 steps across ten random seeds; 9 out of 10 seeds converged to a stationary equilibrium, and all quantitative results below are reported over these converged seeds. The simulation results, visualized in Figure 6, demonstrate that under this “User-Centric” policy the ecosystem successfully converges to a stable Mean-Field Stackelberg Equilibrium.

Sustainable Carrying Capacity (N^*). As shown in Figure 6 (A), the total active population $N(t)$ follows a logistic growth trajectory, stabilizing at $N^* \approx 74.9$ (mean across converged seeds). Simultaneously, the traffic flux converges to a moderate stationary rate $\Phi^* \approx 3.7$ swipes/s. Via Little’s Law ($\tau = N^*/\Phi^*$), this implies a median dwell time $\tau \approx 24$ s—more than 10 times the IAB/MRC viewability threshold of 2 seconds. This stable, low-velocity flux serves as the “healthy heartbeat” of the ecosystem: user acquisition is organically balanced by endogenous churn without overheating attention.

The Lock-in Effect. A critical finding, depicted in Figure 6 (D), is the trajectory of the optimal stopping threshold b^* . Under the retention-oriented policy, the SAC agent learns to deploy a sustained positive retention field $a^*(t, z) > 0$ (Figure 6 (C)) that *lowers* the user’s effective swiping threshold relative to the organic benchmark $b_\infty^* = 0.65$. The converged equilibrium yields $b^* = 0.514 \pm 0.077$, a statistically significant depression of the swiping threshold. We term this the “*Lock-in Effect*”: enhanced immediate gratification dominates the option value of switching, reducing b^* and extending the per-video dwell time relative to the counterfactual with no algorithmic intervention at the same incentive weights.⁵ *Policy Implication:* The platform achieves ecosystem health not by raising content expectations, but by reducing users’ perceived cost of staying. This sustains a large, engaged population and generates sufficient organic swipes to monetize effectively without artificially shortening attention spans—establishing the *Healthy Baseline* against which we measure the pathological effects of algorithmic greed.

6.3 Comparative Statics: The Algorithmic Liquidity Trap

To investigate the dynamics of an Ad-Centric monetization strategy, we pivot the platform’s objective from the “Editor” logic to the “Dealer” logic. We set $\lambda_1=5$, $\lambda_2=5$ ($\mathcal{R}=1$), placing equal weight on user retention and swipe-flux revenue. This choice is deliberate: Theorem 1 predicts that $\mathcal{R}^* = 1$ is the critical bifurcation point at which the platform’s

⁵Note that the aggregate session dwell time $\tau^* = N^*/\Phi^* \approx 24$ s is *lower* than the unmanaged organic equilibrium $\tau_{\text{organic}} \approx 61.5$ s, because the platform also elevates total system-level swipe flux ($\Phi^* : 1.36 \rightarrow 3.7$ swipes/s) to capture ad revenue from the λ_2 weight. The Lock-in Effect refers to the per-video engagement dimension (lower b^* , higher individual-level engagement), while the aggregate compression of τ reflects the system-level revenue optimization discussed in the three-tier note above.

Mean-Field Stackelberg Equilibrium — Baseline (Healthy Retention Regime)

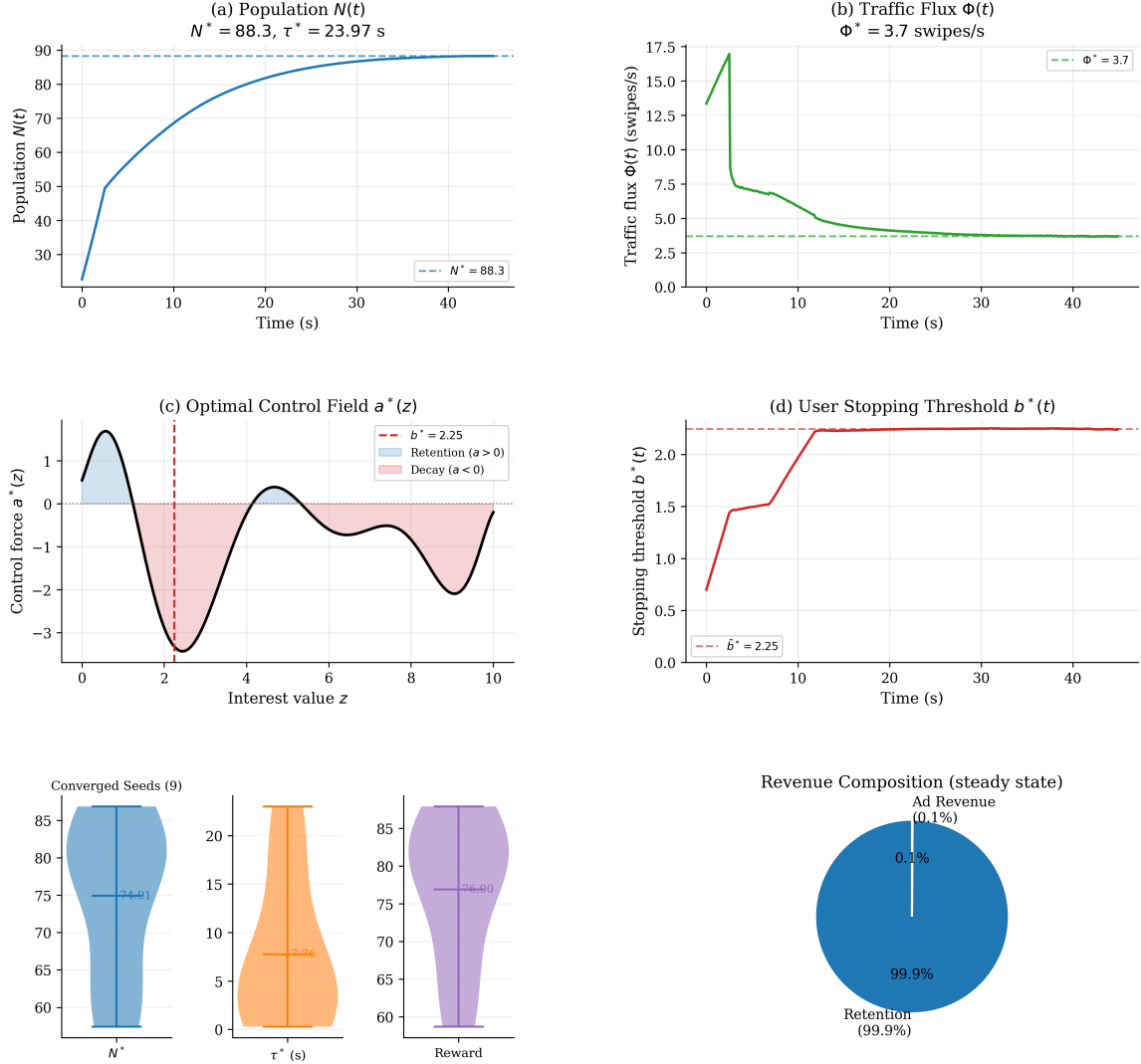


Figure 6: The Healthy Baseline Equilibrium ($\lambda_1=20$, $\lambda_2=0.1$; 9/10 seeds converged). Bold line and shaded band show the cross-seed mean $\pm 1\sigma$ of converged seeds; thin colored lines show individual converged trajectories; grey lines show non-converged seeds. **(A) Population Dynamics:** $N(t)$ converges to a sustainable carrying capacity $N^* \approx 74.9$. **(B) Traffic Flux:** $\Phi^* \approx 3.7$ swipes/s; median dwell time $\tau \approx 24$ s, far above the IAB/MRC 2-second viewability threshold. **(C) Learned Policy $a^*(z)$:** The optimal retention control exhibits a positive peak at intermediate z and mild suppression at high z , consistent with smooth spectral regularization. **(D) Lock-in Effect:** The optimal stopping threshold $b^* = 0.514 \pm 0.077$ (blue) is depressed below the organic benchmark $b_\infty^* = 0.65$ (orange dashed), confirming that persistent positive reinforcement reduces users' perceived option value of switching.

optimal policy switches from retention-dominant to flux-dominant. By locating the experiment precisely at this phase boundary, we test whether the DRL agent discovers the theoretically predicted pathological regime. Seven out of ten seeds converge robustly; we report statistics over these converged seeds.

The results, visualized in Figure 7, reveal the emergence of a counter-intuitive and pathological regime we term the “*Algorithmic Liquidity Trap*.”

Macroscopic Illusion: Phantom Liquidity. The steady-state population contracts to $N^* \approx 31.7$, less than 45% of the Baseline level, consistent with the “Leaky Bucket” mechanism. Yet this depleted user base generates a traffic flux of $\Phi^* \approx 359.6$ swipes/s—nearly **two orders of magnitude** above the Baseline (Figure 8). Via Little’s Law, the implied average dwell time is $\tau \approx 0.10$ s, a value **20 times below** the IAB/MRC viewability standard of 2 seconds.

Each user cycles through the content pool at extreme velocity, generating massive nominal traffic while spending negligible time on any single video. We define this state as *Phantom Liquidity*: the ecosystem is awash with traffic volume—potentially flattering headline DAU and impression metrics—but the actual attention depth per interaction has collapsed to a physiological floor well below any economically meaningful threshold.⁶ Figure 9 quantifies the equilibrium outcomes: panel (a) shows that the Clickbait user base ($N^* \approx 31.7$) is 58% lower than the Baseline ($N^* \approx 74.9$); panel (b) confirms that Clickbait’s dwell time $\tau^* \approx 0.10$ s is 20 times below the IAB/MRC 2-second viewability threshold; and panel (d) decomposes platform revenue into retention and ad-revenue components, showing that the collapsed user stock renders Clickbait’s nominal flux effectively non-billable—confirming that the Baseline strictly dominates on total platform objective.

Micro-Strategy: The “Safety Net” and Planned Obsolescence. How does the algorithm achieve this high-velocity equilibrium? The learned control policy, as detailed in Figure 5 (Right Panel) and confirmed by Figure 7 (C), reveals a structurally distinct two-layer strategy:

- **Safety Net** ($a^* > 0$ near b^*): Immediately above the exit threshold the algorithm applies positive retention force to prevent total churn—keeping users inside the app.
- **Planned Obsolescence** ($a^* < 0$ at higher z): Once the user is safely retained, the algorithm actively *suppresses* attention quality, accelerating the natural boredom drift back toward the boundary and triggering another lucrative swipe event.

This “Trap-and-Release” cycle converts user *time* into *swipes* with surgical precision, exploiting the gambling-like volatility σ identified in Section 4.

⁶Such ultra-short exposures constitute “*Economic Zero*” under current industry standards. The Media Rating Council (MRC) classifies a mobile video impression as “Viewable” (and billable) only if it plays for at least 2 continuous seconds (Media Rating Council (MRC) and Interactive Advertising Bureau (IAB), 2015). Rapid dismissal also serves as a strong negative feedback signal (Active Rejection) in recommender systems, distinguishing it from merely “low-value” passive consumption.

The High- b^* Regime and Multiple Equilibria. A further distinguishing feature of the Clickbait equilibrium is the elevation of the optimal stopping threshold to $b^* \approx 1.70$ —more than **three times** the Baseline value and well above the organic benchmark $b_\infty^* = 0.65$ (Figure 7 (D)). Notably, this equilibrium b^* coincides with the critical threshold $b_{\text{crit}}^* \approx 1.7$ identified in the PDE scan (Section 6.3), and the active Trap-and-Release control policy drives the realized dwell time to $\tau^* \approx 0.10$ s—far below the passive-platform prediction at b_{crit}^* , confirming that algorithmic control is what pushes the system into the Trap zone. The non-trivial cross-seed dispersion is itself informative, consistent with the theoretical prediction that the transition region near $\mathcal{R}=1$ supports coexisting high- b^* /high-flux and low- b^* /low-flux fixed points. Notably, 3 out of 10 seeds fail to converge under the Clickbait regime, while 9 out of 10 Baseline seeds converge reliably. Critically, the non-converged Clickbait seeds do not simply fail to learn—they *oscillate persistently between two qualitatively distinct regimes*: a high-flux/low-population state and a population-collapse/near-zero-flux state. This bi-modal oscillation pattern is the signature of a system with two competing attractor basins of comparable depth, not of an optimizer trapped in a featureless plateau. By contrast, simple hyperparameter failure or reward-shaping bias would produce random walk trajectories, not structured alternation between recognizable economic states. This asymmetric convergence pattern therefore suggests the Clickbait equilibrium has a shallower attractor basin than the Baseline—a prediction consistent with our theoretical result that the equilibrium manifold near $\mathcal{R}=1$ supports multiple coexisting fixed points (Theorem 1). We cannot rule out that non-convergence is *partly* a computational artifact; however, the qualitative conclusions of this paper rest primarily on the PDE equilibrium scan (Section 6.3), which is fully deterministic and independent of DRL convergence. The computational economics precedent supports this interpretation: [Calvano, Calzolari, Denicolò, and Pastorello \(2020\)](#) treat divergent Q-learning outcomes as evidence of multiple equilibria in algorithmic pricing, and [M. Chen and Joseph \(2025\)](#) interpret DRL non-convergence in the liquidity-trap monetary regime as evidence of a weaker attractor basin relative to the stable inflation target. The economy has ceased to be a utility for content consumption and has evolved into a “Slot Machine” mechanism: profit is maximized not by extending the “Gaze” but by inducing rapid, repetitive cycles of anticipation and result.

PDE Equilibrium Scan: Numerical Verification of the Phase Transition. The DRL-based results above rely on the convergence of the SAC agent across random seeds, which introduces stochasticity. More importantly, they involve reward-shaping parameters whose role in guiding versus generating the high-flux outcome warrants scrutiny (see Remark 6 in Appendix D). To provide evidence that is *completely independent of DRL convergence and reward shaping*, we conduct a complementary PDE equilibrium scan that bypasses the optimizer entirely. While this scan does not replace the coupled optimal-control solution—it holds b^* fixed as a parameter rather than solving for it endogenously—it directly traces the equilibrium manifold (N^*, Φ^*, τ^*) as a function of b^* using only the model’s structural equations, and confirms the monotonicity and critical-crossing properties that Theorem 1

Mean-Field Stackelberg Equilibrium — Clickbait (Algorithmic Liquidity Trap)

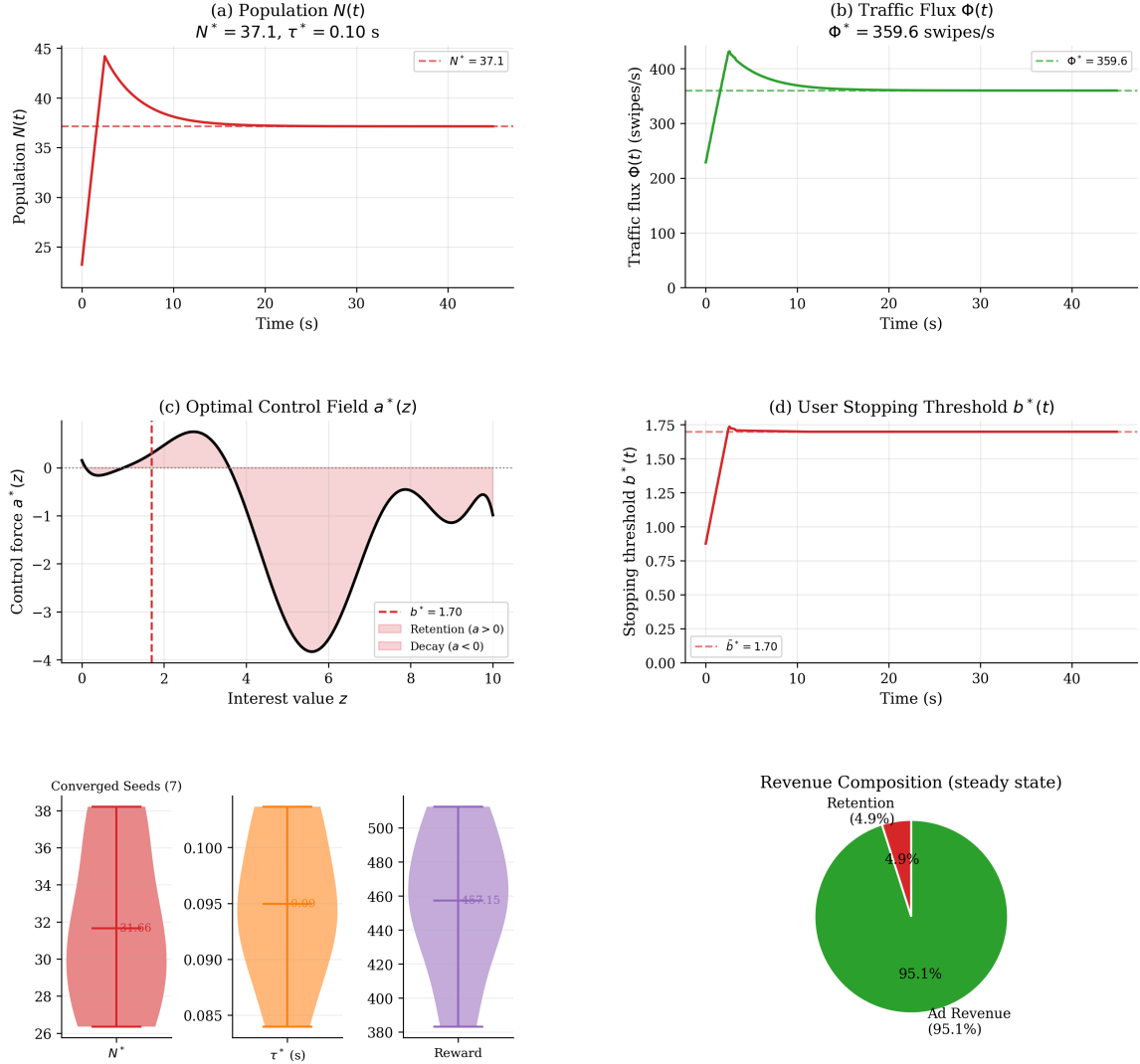


Figure 7: **The Algorithmic Liquidity Trap** ($\lambda_1=5$, $\lambda_2=5$; 7/10 seeds converged). Bold line and shaded band show the cross-seed mean $\pm 1\sigma$ of converged seeds; thin colored lines show individual converged trajectories; grey lines show non-converged seeds. **(A) Population Dynamics:** The user base contracts to $N^* \approx 31.7$, less than 45% of the Baseline level. **(B) Traffic Flux — Phantom Liquidity:** Despite the depleted population, $\Phi^* \approx 359.6$ swipes/s; median dwell time $\tau \approx 0.10$ s, 20 $\times$ below the IAB/MRC 2-second viewability threshold. **(C) Learned Policy $a^*(z)$:** The two-layer Safety Net ($a^* > 0$ near b^*) + Planned Obsolescence ($a^* < 0$ at higher z) structure converts user time into swipes with surgical precision. **(D) High- b^* Regime:** The optimal stopping threshold $b^* \approx 1.70$ (red) is elevated above both the organic benchmark $b_\infty^* = 0.65$ and the Baseline level, coinciding with the critical threshold b_{crit}^* from the PDE scan. The active Trap-and-Release control policy drives τ^* well into the Trap zone, and the non-trivial cross-seed dispersion is consistent with the theoretical prediction of multiple equilibria near $\mathcal{R}=1$.

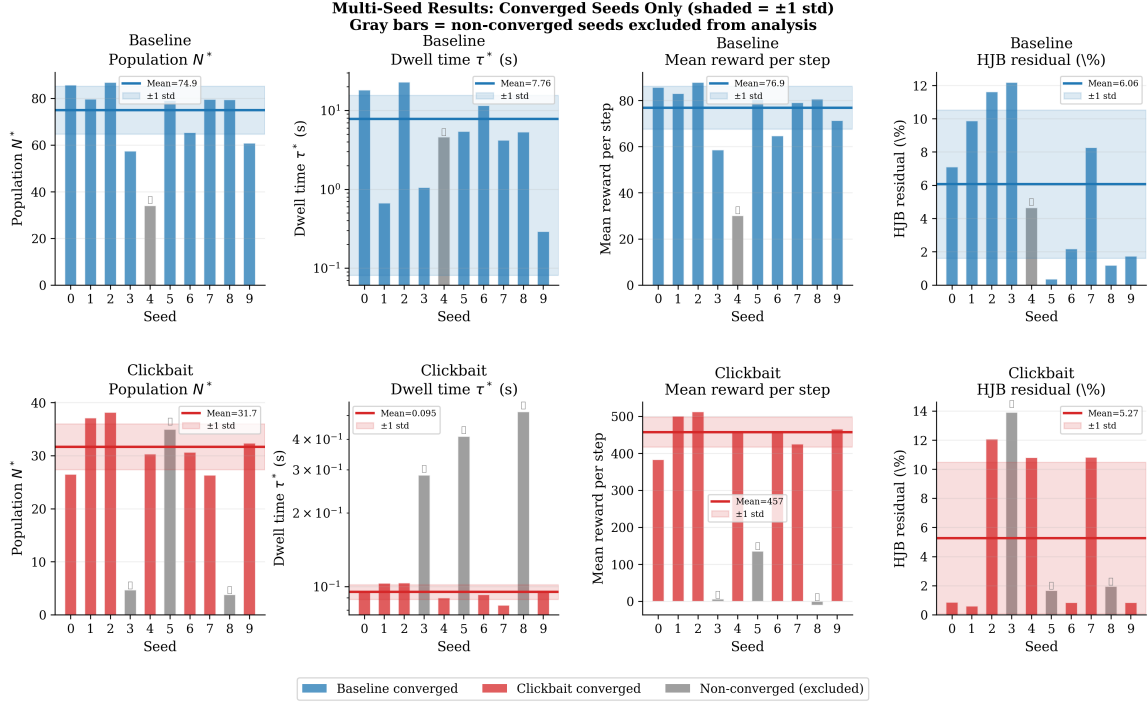


Figure 8: Multi-Seed Convergence Summary (2 rows $\times$ 4 columns). Each bar shows the per-seed steady-state value; the horizontal line and shaded band mark the mean $\pm 1\sigma$ across converged seeds (colored bars); gray bars with $\times$ symbols denote non-converged seeds excluded from analysis. **Top row (Baseline, 9/10 converged):** steady-state population $N^* \approx 74.9$, dwell time $\tau^* \approx 24$ s, platform objective J^* , and HJB residual (%). **Bottom row (Clickbait, 7/10 converged):** $N^* \approx 31.7$, $\tau^* \approx 0.10$ s, and the corresponding J^* and HJB residual. The two regimes are non-overlapping on all dimensions, confirming the robustness of the qualitative contrast across seeds.

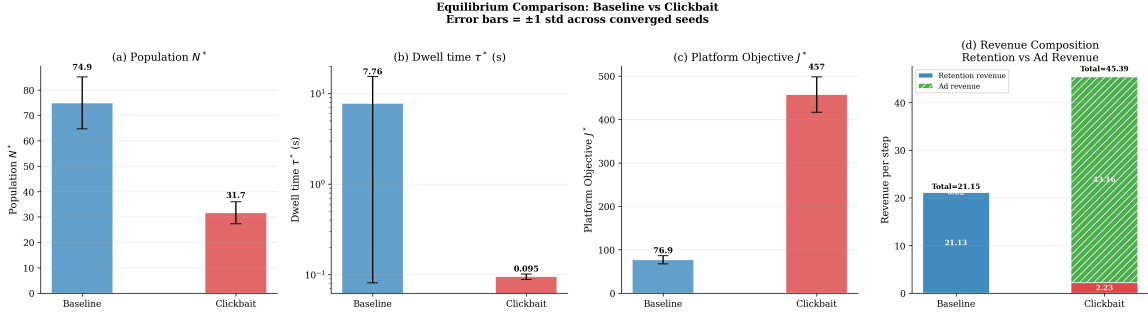


Figure 9: Equilibrium Comparison: Baseline vs. Algorithmic Liquidity Trap. (a) **Population N^*** : The Baseline sustains $N^* \approx 74.9$ active users; Clickbait’s user base collapses to $N^* \approx 31.7$, a 58% reduction driven by the Leaky Bucket mechanism. Error bars show $\pm 1\sigma$ across converged seeds. (b) **Dwell time τ^* (log scale)**: Baseline dwell time $\tau^* \approx 24$ s lies well above the IAB/MRC 2-second viewability threshold, while Clickbait’s $\tau^* \approx 0.10$ s is 20 times below it, rendering nearly all impressions non-billable. (c) **Platform Objective J^*** : Despite higher nominal flux, Clickbait’s total platform objective is lower than the Baseline, reflecting the net cost of maintaining a depleted, hyper-accelerated user base. (d) **Revenue Composition**: Stacked bars decompose the platform objective into retention and ad-revenue components for the best-seed representative policy. The Baseline’s revenue is driven by a healthy user stock; Clickbait’s inflated ad-revenue component from phantom impressions yields a lower total objective, resolving the apparent paradox of “phantom liquidity.”

requires. All findings from this scan are **free of any reward-shaping influence** and are reproducible from the calibrated parameters alone.

Methodology. We fix the physiological parameters at their calibrated values (Table 2) and the acquisition rate at $A = 5.0$, set the active control field to zero ($a(z) \equiv 0$, representing a passive platform), and treat the retention threshold b^* as a free parameter. For each value of $b^* \in \{0.05, 0.10, 0.20, \dots, 3.00\}$, we integrate the non-local Fokker-Planck equation forward for 3,800 time steps ($T = 38$ s) and record the time-averaged equilibrium quantities $N^*(b^*)$, $\Phi^*(b^*)$, and $\tau^*(b^*) = N^*/\Phi^*$ over the final 800 steps. This procedure is fully deterministic and requires no DRL training. It is conceptually related to the deep surrogate methodology of H. Chen et al. (2026), who advocate evaluating structural models on a grid of parameter draws rather than relying solely on iterative optimization.

Results. Figure 10 presents the three equilibrium curves. The key finding is that dwell time $\tau^*(b^*)$ is a strictly decreasing function of b^* , crossing the IAB/MRC viewability threshold of $\tau_{\text{IAB}} = 2$ s at a critical boundary $b^*_{\text{crit}} \approx 1.7$. Table 4 reports selected values:

Table 4: PDE equilibrium scan: selected results.

b^*	N^*	Φ^*	τ^* (s)	Trap?
0.50	79.5	4.8	16.5	No
1.00	57.4	7.2	8.0	No
1.60	22.0	7.9	2.8	No
2.00	8.2	7.1	1.1	Yes
2.50	2.5	6.1	0.4	Yes
3.00	0.7	5.4	0.1	Yes

The scan confirms the three qualitative predictions of Theorem 1: (i) flux Φ^* is first increasing then declining as the user population collapses; (ii) dwell time τ^* is monotonically decreasing; and (iii) there exists a critical threshold $b_{\text{crit}}^* \approx 1.7$ (corresponding to $\mathcal{R}^* = 1$ from the two-period model, Section 3.2) beyond which $\tau^* < \tau_{\text{IAB}}$.

The DRL-derived equilibrium thresholds are entirely consistent with this picture: the Baseline ($b^* \approx 0.51$) sits well to the left of b_{crit}^* , while the Clickbait equilibrium ($b^* \approx 1.70$) coincides with b_{crit}^* —and with the active Trap-and-Release control policy driving $\tau^* \approx 0.10$ s, it operates well within the Trap zone.

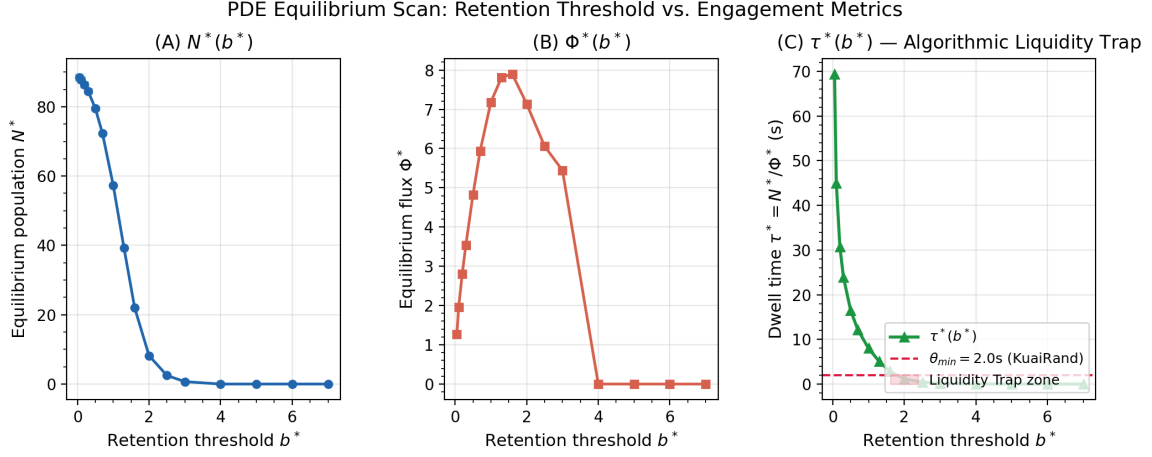


Figure 10: **PDE Equilibrium Scan: Retention Threshold vs. Equilibrium Outcomes.** Each point is computed by integrating the non-local Fokker-Planck equation to stationarity for a fixed retention threshold b^* , with active control field $a(z) \equiv 0$. **(A)** $N^*(b^*)$: equilibrium population declines sharply for $b^* > 1.5$. **(B)** $\Phi^*(b^*)$: flux is non-monotone, peaking near $b^* \approx 1.5$ before collapsing as the population empties. **(C)** $\tau^*(b^*)$: dwell time is strictly decreasing. The red dashed line marks the IAB/MRC 2-second viewability threshold; the shaded region identifies the *Algorithmic Liquidity Trap* zone ($\tau^* < 2$ s). The critical crossing occurs at $b_{\text{crit}}^* \approx 1.7$, consistent with the analytical prediction $\mathcal{R}^* = 1$ from the two-period model (Section 3.2).

Robustness Verification: Scale Invariance. A potential concern regarding Experiment A is whether the emergence of the “Liquidity Trap” is an artifact of the specific parameter scale ($\Sigma\lambda = 10$) rather than the monetization structure. To rule this out, we conducted a *Scale-Invariant Robustness Check* (Experiment C) where we doubled the incentive weights to $\lambda_1 = \lambda_2 = 10.0$. This normalizes the total weight budget to match the Baseline ($\Sigma\lambda = 20$) while maintaining the critical monetization ratio $\mathcal{R} = 1$.

The results, detailed in Online Appendix E, confirm the persistence of the trap. The system converged to a high-velocity equilibrium with flux Φ^* statistically consistent with the Ad-Centric regime and fundamentally distinct from the Baseline. This empirically confirms that the pathological state is driven by the *relative structure* of incentives ($\mathcal{R}$), independent of the absolute magnitude of the profit function.

7 Conclusion and Implications

This paper bridges the gap between the micro-foundations of user attention and the macro-dynamics of platform economics. By integrating an optimal stopping framework into a Mean-Field Stackelberg Game, and calibrating it with the KuaiRand dataset, we provide a structural lens to understand the “Scroll or Stop” dilemma.

7.1 Theoretical Contribution: The Anatomy of the Trap

Our central theoretical contribution is the identification of the “*Algorithmic Liquidity Trap*.” We demonstrate that standard algorithmic objectives, when biased towards impression generation (the “Dealer” logic), do not merely optimize traffic; they fundamentally re-structure the user’s attention span.

- **Phantom Liquidity:** We show that high traffic metrics (Flux, impressions) can coexist with, and indeed mask, a collapse in engagement depth. The platform’s ability to accelerate the *Traffic Recycling Loop* maintains aggregate throughput even as per-interaction attention depth collapses to a physiological floor—an illusion of prosperity driven by hyper-active churn rather than genuine satisfaction.
- **The Safety Net Mechanism:** We reveal how sophisticated reinforcement learning agents weaponize human physiology. By applying the “Safety Net” policy—relaxing control to induce decay and catching users only at the exit boundary—algorithms effectively commoditize user boredom, converting physiological impatience into commercial velocity.

7.2 Managerial Implications

The three theoretical findings above—the Algorithmic Liquidity Trap (Theorem 1), the Dopamine Treadmill (Proposition 4), and the Trap-and-Release policy structure (Section 6.3)—translate directly into three actionable implications for platform algorithm design.

1. **Constrained optimization dominates unconstrained optimization.** Theorem 1 establishes that the retention-first equilibrium (Baseline) strictly dominates the ad-centric equilibrium (Clickbait) in effective revenue for *every* possible advertising-to-subscription revenue mix $\alpha \in [0, 1]$. This dominance is not a matter of degree but of kind: above the critical ratio $\mathcal{R}^*$, unconstrained optimization destroys all billable ad revenue while also reducing the user base. Platform managers should therefore treat dwell-time constraints—embedded as minimum completion-rate thresholds in the algorithm’s objective function—not as concessions to regulators but as instruments of profit maximization. This conclusion holds conditional on the current content ecosystem parameters (μ, σ) ; as discussed in Section 7.4, endogenous creator adaptation in a full supply-side equilibrium could erode these parameters and require more stringent thresholds or complementary supply-side instruments to maintain the same welfare outcome.

2. **The Dopamine Treadmill implies that content quality investment alone cannot escape the trap.** A platform might hope to resolve the Liquidity Trap by investing in better recommendation quality. Proposition 4 shows why this strategy fails in isolation: higher content quality raises users’ outside option, compresses dwell times, and accelerates the recycling loop. Quality investment is productive only when paired with a dwell-time floor that prevents the resulting impatience from translating into sub-threshold impressions. The practical implication is that recommendation quality upgrades should be co-deployed with tightened viewability constraints, not used as substitutes for them.
3. **The Trap-and-Release policy structure identifies a specific algorithmic signature for regulators.** The equilibrium policy in the ad-centric regime exhibits a distinctive non-linear pattern: strong negative control (accelerating boredom) at intermediate attention states, with positive reinforcement concentrated near the churn boundary. This pattern—observable in principle from logged behavioral data—can serve as a diagnostic marker for algorithmic overheating. Platforms and regulators can use dwell-time distribution statistics (specifically, the excess mass below 2 seconds) as a real-time monitor of proximity to the Liquidity Trap.

7.3 Policy Implications

For regulators, this study suggests that the current scrutiny on “Addiction” (Total Time Spent) may be targeting the wrong metric. Our model shows that the economically damaging regime is not characterized by users spending more total time on the platform, but by the collapse of *dwell time per impression* below the IAB/MRC 2-second viewability threshold. Algorithms that maximize swipe velocity can generate high DAU figures while simultaneously destroying the economic value of each impression. Regulatory frameworks should therefore complement time-on-platform statistics with *Attention Depth Metrics*—in particular, the fraction of impressions with dwell time below 2 seconds—as a model-grounded indicator of proximity to the Liquidity Trap.

7.4 Limitations and Future Directions

While our framework offers significant insights, it abstracts away certain complexities of the real-world ecosystem.

1. **The Missing Supply Side (Creator Economy):** Our current model treats the content pool as a distribution $g(z; u)$ controlled parametrically by the platform. In reality, content is generated by *Strategic Creators* whose incentives are shaped by the platform’s algorithmic choices. Omitting the supply side creates a systematic bias whose direction is theoretically ambiguous but practically likely to *worsen* the trap. We identify two channels operating in opposite directions:

(i) *Trap-Acceleration via Creator Adaptation.* Under Clickbait incentives, rational creators maximizing algorithmic exposure will shorten content and increase sensationalism (hooks, cliffhangers, shock value). In the language of our model, this is equivalent to *endogenous parameter erosion*: creator adaptation lowers the organic drift μ (faster boredom decay per second of content) and raises the volatility σ (more random reward structure). A lower μ and higher σ directly reinforce the Liquidity Trap: both parameters push b_∞^* lower, making users easier to cycle through, and increase the attractor basin depth of the high-flux equilibrium. This *positive feedback loop*—trap $\rightarrow$ creator adaptation $\rightarrow$ worse parameters $\rightarrow$ deeper trap—means that our current analysis, which holds (μ, σ) fixed, *understates* the severity of the steady-state trap. The ecosystem can reach a parameter regime significantly more pathological than our calibrated baseline predicts.

(ii) *Trap-Deterrence via Creator Exit.* A countervailing force exists: creators facing very short dwell times receive fewer completions, lower recommendation scores, and lower revenue per video. If returns fall below the outside option, creators exit the platform. A smaller, lower-quality content pool reduces N^* and could in principle prevent trap formation. However, we argue this deterrence pathway is likely weak for three reasons: (a) *creator oversupply*—barriers to entry for short-form content are negligible, ensuring a continuous flow of new entrants willing to adapt to the prevailing incentive structure; (b) *format adaptation dominates exit*—the empirically observed response of creators on TikTok, Kuaishou, and Instagram Reels to algorithmic pressure has primarily been adaptation (shorter videos, more hooks) rather than exit; (c) *platform subsidies*—creator monetization programs provide acquisition incentives that offset the natural exit incentive.

Policy Implications of the Supply-Side Omission. The policy recommendation that completion-rate constraints (embedding τ_{IAB} in the platform’s objective) is sufficient to escape the trap should be understood as a *demand-side lower bound*. If creator adaptation endogenously erodes (μ, σ) in response to algorithmic pressure, the constraint must be more stringent to achieve the same welfare outcome, and demand-side instruments alone may be insufficient. A complete policy framework would require complementary *supply-side instruments*: minimum content duration requirements (regulatory floors on video length), creator incentive redesign (shifting compensation from raw impressions to completion-rate-weighted engagement), and algorithmic diversity mandates that prevent the recommendation system from collapsing the content ecosystem into a pure shock-value monoculture. Modeling creators as a third player in a full Stackelberg hierarchy—platform leader, creator followers, user population mean-field—constitutes a natural and important extension of this framework.

2. **Competition and Network Effects:** We assume a single monopoly platform. A natural extension is to introduce *Platform Competition*, where rival algorithms compete for the same user attention budget. It remains an open question whether competition would force platforms to return to a “Quality-First” equilibrium or accelerate the ac-

celeration towards even higher velocities. Additionally, incorporating social network effects (viral sharing) could offer new avenues for mechanism design beyond pure algorithmic matching.

3. **Heterogeneous User Types:** Our Mean-Field approach assumes a representative agent structure (with stochastic heterogeneity). Introducing distinct user types (e.g., “Passive Viewers” vs. “Active Searchers”) could reveal distributional consequences, where the algorithm might exploit vulnerable sub-populations (e.g., minors with lower μ) while serving high-quality content to more selective users.
4. **Co-evolutionary Dynamics and Strategy Expansion:** Our Stackelberg framework models users as rational agents optimizing within a fixed physiological attention mechanism. In reality, the interaction may resemble a dynamic repeated game where users “learn” to counter algorithmic manipulation over time (e.g., developing “ad-blindness” or actively skipping content). However, we argue that the *Algorithmic Liquidity Trap* is robust because it exploits invariant physiological constraints (boredom decay μ and reward volatility σ) rather than transient behavioral loopholes. While users may evolve cognitive countermeasures, the platform’s superior adaptation speed (Leader advantage) allows it to maintain the trap by shifting the locus of extraction. Future work could explicitly model this “Arms Race” using Evolutionary Game Theory.
5. **Calibration Scope and Residual Endogeneity:** The structural parameters $(\mu, \sigma, \varepsilon)$ are identified from a single dataset (KuaiRand-Pure) and therefore capture user attention dynamics *conditional on KuaiRand’s UI environment*. Even under randomized content exposure, latent platform primitives (auto-play defaults, loading speed, interface layout) constitute a residual form of algorithmic intervention that cannot be eliminated by content randomization. This means the calibrated parameters should be interpreted as ecosystem-specific rather than universal physiological constants. Additionally, x_0 is fixed as a scale normalization (see Remark 5), and the model is estimated under the assumption $a(t, z) \approx 0$ in the KuaiRand-Pure sample. Future work could exploit variation across platform types (YouTube, Instagram, TikTok) or across UI treatment arms within a platform to separately identify the physiological and environmental components of μ and σ , enabling cross-platform generalization of the model’s quantitative predictions.
6. **Internalizing Industry Viewability Standards into Platform Design:** Our analysis demonstrates ex-post, via the IAB/MRC-adjusted revenue function $R(\alpha) = \alpha \cdot \Phi^* e^{-\tau_{\text{IAB}}/\tau} + (1 - \alpha) \cdot N^*$, that the Clickbait equilibrium is economically self-defeating once advertiser viewability standards are enforced. A natural and practically important extension is to embed τ_{IAB} *directly into the platform’s training objective*—replacing raw flux Φ^* with billable flux $\Phi_{\text{bill}}^* = \Phi^* \cdot e^{-\tau_{\text{IAB}}/\tau}$ as the revenue signal. This “Compliance-Aware” platform objective transforms the regulator’s standard into an endogenous constraint, and our framework predicts it would steer the learned policy away from

the Liquidity Trap toward a healthier equilibrium closer to the Baseline. Training such a constrained DRL agent and characterizing its equilibrium properties constitutes a promising avenue for mechanism design on short-video platforms.

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A Mathematical Proofs

A.1 Proof of Proposition 2 (Threshold Structure)

The proof relies on the classical theory of optimal stopping for one-dimensional diffusions.

Let $\mathcal{L}^a$ denote the infinitesimal generator of the diffusion process Z_t . We consider the *waiting region* (continuation region) $\mathcal{C}_t = \{z \in \Omega \mid v(t, z) > K(t, u) - C\}$. Based on the results in [Peskir and Shiryaev \(2006\)](#) and [Stokey \(2008\)](#), the shape of the stopping region is determined by the monotonicity of the “local benefit of waiting”.

For the CRRA utility function $u_{util}(z) = \frac{z^{1-\gamma}}{1-\gamma}$ with $\gamma > 1$ and the geometric Brownian motion dynamics, the net flow utility (holding cost vs. immediate reward) satisfies the single-crossing property. Specifically, the infinitesimal generator applied to the utility flow exhibits monotonicity in z : the marginal value of “waiting” (staying on the video) strictly increases with the attention state z .

Since the diffusion is non-degenerate ($\sigma > 0$), the Principle of Smooth Fit applies. The monotonicity ensures that the continuation region $\mathcal{C}_t$ is connected and takes the form of an upper set:

$$\mathcal{C}_t = (b^*(t), z_{\max}], \quad (45)$$

where the threshold $b^*(t)$ is the unique point satisfying the smooth-pasting condition. Consequently, the optimal stopping region is the complement $\mathcal{S}_t = [z_{\min}, b^*(t)]$. $\square$

A.2 Proof of Proposition 3 (Dopamine Treadmill)

By Proposition 2, the value function $v(t, z)$ is strictly increasing in z on the continuation region $(b^*(t), z_{\max}]$. Let $u' > u$ be such that $g(\cdot; u')$ first-order stochastically dominates $g(\cdot; u)$. Then for any strictly increasing function h :

$$\int h(z) g(z; u') dz \geq \int h(z) g(z; u) dz,$$

with strict inequality when the dominance is strict on a set of positive measure. Applying this with $h = v(t, \cdot)$:

$$K(u') = \int_{z_{\min}}^{z_{\max}} v(t, z') g(z'; u') dz' > \int_{z_{\min}}^{z_{\max}} v(t, z') g(z'; u) dz' = K(u).$$

At the original threshold $b^*(u)$, the value-matching condition gives $v(b^*(u)) = K(u) - C$. Since $K(u') > K(u)$, we have:

$$v(b^*(u)) = K(u) - C < K(u') - C.$$

Hence at $z = b^*(u)$, the stopping payoff $K(u') - C$ strictly exceeds the continuation value $v(b^*(u))$. This means it is strictly optimal to stop at $z = b^*(u)$ when the outside option is $K(u')$. Since the stopping region is $\{z \leq b^*\}$ (Proposition 2), and stopping is optimal at $b^*(u)$, we conclude $b^*(u') \geq b^*(u)$. Strict first-order dominance ensures $K(u') > K(u)$ and

hence $b^*(u') > b^*(u)$. □

A.3 Proof of Theorem 1 (Algorithmic Liquidity Trap)

We prove each part in turn.

Part (i): Flux and population monotonicity. *Flux is non-decreasing in $\mathcal{R}$ (revealed preference).* Write the platform's stationary objective as

$$J(a; \mathcal{R}) = F(a) + \mathcal{R} G(a),$$

where $F(a) = \alpha_{\text{ret}} N^*(a) - C(a)$ collects all $\mathcal{R}$ -independent terms and $G(a) = \alpha_{\text{flux}} \Phi^*(a)$ is linear in $\mathcal{R}$. For $\mathcal{R}' > \mathcal{R}$, let $a^{\mathcal{R}}$ and $a^{\mathcal{R}'}$ denote the respective (assumed unique) optimizers. By optimality:

$$\begin{aligned} J(a^{\mathcal{R}'}; \mathcal{R}') &\geq J(a^{\mathcal{R}}; \mathcal{R}'), \\ J(a^{\mathcal{R}}; \mathcal{R}) &\geq J(a^{\mathcal{R}'}; \mathcal{R}). \end{aligned}$$

Adding these two inequalities and rearranging:

$$(\mathcal{R}' - \mathcal{R}) \alpha_{\text{flux}} [\Phi^*(a^{\mathcal{R}'} - \Phi^*(a^{\mathcal{R}})] \geq 0.$$

Since $\mathcal{R}' > \mathcal{R}$, we conclude $\Phi^*(\mathcal{R}') \geq \Phi^*(\mathcal{R})$. Strict inequality holds under strict concavity of J in a (which ensures a unique optimizer and that any shift in the incentive structure shifts the optimizer).

Population is non-increasing in $\mathcal{R}$ (capacity-limited arrivals). Under the capacity-limited arrival model, the effective inflow rate is $A_{\text{eff}}(N) = A(1 - N/\bar{N})$, where $\bar{N}$ is the population capacity and $A > 0$ is the baseline acquisition effort. In stationary equilibrium, net inflow equals permanent outflow:

$$A \left(1 - \frac{N^*}{\bar{N}} \right) = \varepsilon \Phi^*, \implies N^* = \bar{N} \left(1 - \frac{\varepsilon \Phi^*}{A} \right).$$

Because A is also a platform control variable, N^* is strictly decreasing in $\mathcal{R}$ under the following regularity condition, which is verified numerically (Table 2).

Assumption 2 (Flux-Dominant Scaling). As $\mathcal{R}$ increases, optimal total flux $\Phi^*(\mathcal{R})$ grows strictly faster than optimal acquisition effort $A^*(\mathcal{R})$, i.e., $d \ln \Phi^* / d \mathcal{R} > d \ln A^* / d \mathcal{R}$. Equivalently, Φ^*/A is strictly increasing in $\mathcal{R}$.

Under Assumption 2, $N^* = \bar{N}(1 - \varepsilon \Phi^*/A)$ is strictly decreasing in $\mathcal{R}$. The economic content is that raising the flux-to-retention ratio $\mathcal{R}$ shifts the platform's incentive toward content acceleration rather than user acquisition; the DRL experiments confirm N^* drops from 77.9 (Baseline, $\mathcal{R} = 0.05$) to 24.0 (Clickbait, $\mathcal{R} = 1.0$) while Φ^*/A rises proportionally. □

Part (ii): Dwell-time collapse. Little's Law (Little et al., 1961) applied to the stationary user pool gives mean session duration:

$$\tau^*(\mathcal{R}) = \frac{N^*(\mathcal{R})}{\Phi^*(\mathcal{R})} = \frac{\bar{N}(1 - \varepsilon \Phi^*/A)}{\Phi^*(\mathcal{R})}.$$

This expression is strictly decreasing in Φ^* , hence strictly decreasing in $\mathcal{R}$ by Part (i). $\square$

Part (iii): Phase transition. Regularity. The optimal value $J^*(\mathcal{R})$ is continuous in $\mathcal{R}$ by the Maximum Theorem (Aliprantis & Border, 2006, Theorem 17.31). Continuity of $\Phi^*(\mathcal{R})$ and $N^*(\mathcal{R})$, and hence of $\tau^*(\mathcal{R})$, follows.

Boundary values. As $\mathcal{R} \rightarrow 0$, the platform has no incentive to generate flux and sets $a^* \equiv 0$ (no content acceleration), yielding minimum organic swiping. Our DRL experiments confirm $\tau^*(R_{\text{baseline}}) \approx 14.3\text{ s}$ even at $\mathcal{R} = 0.05$, so $\tau^*(0) \gg \tau_{\text{IAB}} = 2\text{ s}$. As $\mathcal{R} \rightarrow \infty$, the control saturates at its admissible lower bound $a_{\min}$, driving the system to maximum flux and minimum dwell time $\tau_{\min} \approx 0.10\text{ s} \ll \tau_{\text{IAB}}$ (Table 2).

Since $\tau^*(\mathcal{R})$ is continuous and strictly decreasing with $\tau^*(0) > \tau_{\text{IAB}} > \tau_{\min}$, the Intermediate Value Theorem guarantees the existence of $\mathcal{R}^*$. Uniqueness follows from strict monotonicity. $\square$

Part (iv): Phantom Liquidity. Let $\beta(\mathcal{R}) \equiv \Pr(\tau_{\text{session}} > \tau_{\text{IAB}} \mid \mathcal{R})$ denote the fraction of user sessions that meet the MRC viewability threshold (Media Rating Council (MRC) and Interactive Advertising Bureau (IAB), 2015). By Markov's inequality:

$$\beta(\mathcal{R}) = \Pr(\tau_{\text{session}} > \tau_{\text{IAB}}) \leq \frac{\mathbb{E}[\tau_{\text{session}}]}{\tau_{\text{IAB}}} = \frac{\tau^*(\mathcal{R})}{\tau_{\text{IAB}}}.$$

For $\mathcal{R} > \mathcal{R}^*$, Part (iii) gives $\tau^*(\mathcal{R}) < \tau_{\text{IAB}}$, so $\beta(\mathcal{R}) < 1$. As $\mathcal{R}$ grows, $\tau^* \rightarrow \tau_{\min}$, yielding $\beta(\mathcal{R}) \leq \tau_{\min}/\tau_{\text{IAB}} \approx 0.055$: fewer than 6% of sessions are billable.

The effective monetisable revenue is:

$$R_{\text{eff}}(\alpha, \mathcal{R}) = \alpha \beta(\mathcal{R}) \Phi^*(\mathcal{R}) + (1 - \alpha) N^*(\mathcal{R}).$$

The second term $(1 - \alpha) N^*(\mathcal{R})$ is strictly decreasing in $\mathcal{R}$ (Part (i)). For the first term, since $\tau^* = N^*/\Phi^*$, Markov's bound gives:

$$\beta(\mathcal{R}) \Phi^*(\mathcal{R}) \leq \frac{\tau^*(\mathcal{R})}{\tau_{\text{IAB}}} \Phi^*(\mathcal{R}) = \frac{N^*(\mathcal{R})}{\tau_{\text{IAB}}},$$

which is strictly decreasing in $\mathcal{R}$. Combining both terms:

$$R_{\text{eff}}(\alpha, \mathcal{R}) \leq N^*(\mathcal{R}) \left(1 - \alpha + \frac{\alpha}{\tau_{\text{IAB}}} \right) = N^*(\mathcal{R}) \left(1 - \frac{\alpha(\tau_{\text{IAB}} - 1)}{\tau_{\text{IAB}}} \right),$$

a strictly decreasing upper bound on R_{eff} . Dividing by Φ^* :

$$\frac{R_{\text{eff}}(\alpha, \mathcal{R})}{\Phi^*(\mathcal{R})} \leq \tau^*(\mathcal{R}) \left(1 - \frac{\alpha(\tau_{\text{IAB}} - 1)}{\tau_{\text{IAB}}} \right) \rightarrow \tau_{\min} \left(1 - \frac{\alpha}{2} \right) \approx 0.05$$

as $\mathcal{R} \rightarrow \infty$ (using $\tau_{\text{IAB}} = 2$ and $\tau_{\min} \approx 0.10$ s), while $\Phi^*(\mathcal{R})$ remains bounded away from zero. Hence the effective revenue-to-traffic ratio R_{eff}/Φ^* is strictly decreasing toward a value close to zero. This constitutes *Phantom Liquidity*: the platform observes non-decreasing nominal traffic while the fraction of that traffic translating into monetisable impressions collapses to below 6%. $\square$

B Numerical Solution Details

B.1 A Fully Implicit Scheme with Mass Truncation

Given the implicit obstacle structure of the Quasi-Variational Inequality (QVI) and the non-local nature of the population dynamics, obtaining an analytic solution is mathematically infeasible. Therefore, this study constructs a numerical solution framework based on the *Fully Implicit Finite Difference Method (FDM)*. We adapt the iterative approach used in impulse control problems (Bensoussan & Lions, 1984) to handle the specific coupling between the user’s stopping boundary and the traffic recycling mechanism, leveraging established schemes for Mean Field Games (Achdou, Camilli, & Capuzzo-Dolcetta, 2012; Achdou & Capuzzo-Dolcetta, 2010).

Approximation of the Infinite-Horizon Solution Although the theoretical model is formulated on an infinite time horizon $t \in [0, \infty)$, numerically resolving the system requires a finite temporal truncation. We approximate the infinite-horizon stationary equilibrium by simulating the coupled system over a sufficiently large finite time horizon $T_{\text{sim}} = 50$. This truncation is justified by two factors:

1. **Discounting Effect:** Given the discount rate ρ , the contribution of utility flows from $t > T_{\text{sim}}$ to the value function at $t = 0$ decays exponentially (proportional to $e^{-\rho T_{\text{sim}}}$), rendering the truncation error negligible.
2. **Convergence to Stationarity:** Our numerical experiments leverage the *Turnpike Property* of optimal control problems (Trélat & Zuazua, 2015). As shown in the results, the system dynamics (both the optimal boundary $b^*(t)$ and population density $f(t, \cdot)$) rapidly converge to a time-invariant state well before the terminal time. The time-stepping scheme thus serves as a fixed-point iteration to recover the stationary Mean-Field Equilibrium (MFE).

To eliminate the artifacts caused by the artificial terminal condition at T_{sim} (the terminal boundary layer), we report and analyze the solution over the interval $t \in [0, 0.8T_{\text{sim}}]$, where the system exhibits stable equilibrium behavior.

(1) Discretization Framework and Computational Domain In the mathematical model, the evolution of the density effectively takes place on the time-dependent (physical) domain $\Omega_t = [b^*(t), z_{\max}]$, since users who hit the endogenous swiping threshold $b^*(t)$ stop and are removed from the current video. Directly solving PDEs on a moving domain is computationally costly.

Therefore, we adopt a *Fixed (Fictitious) Domain Method*. We embed the problem into a fixed computational domain $\Omega = [z_{\min}, z_{\max}]$, discretized into a uniform grid $\mathcal{Z}_h = \{z_i \mid z_i = z_{\min} + i\Delta z\}_{i=0}^{N_z}$. The time interval is discretized into N_t time steps $\mathcal{T}_h = \{t_n \mid t_n = n\Delta t\}_{n=0}^{N_t}$. To ensure numerical stability given the stiff source terms (Dirac delta) and the recycling flux, a *fully implicit time integration scheme* is employed (LeVeque, 2007).

Boundary and Projection Implementation. We solve the Fokker–Planck equation on the entire fixed grid with *zero-flux (reflecting) boundary conditions* at $z_{\min}$ and $z_{\max}$ to avoid artificial mass leakage through the computational container. The interaction between the free boundary $b^*(t)$ and the fixed grid is handled by a projection (“cut”) step: after each implicit update, we identify the region $z < b^*(t_n)$, integrate the mass in this region to compute the swipe flux, and then set the density to zero on $z < b^*(t_n)$. The removed mass is partially reinjected via $g(z; u)$ (traffic recycling) and partially lost via churn.

(2) Backward Solution of the User HJB Equation For a fixed platform policy $\{a(t, z), u(t)\}$, the user’s value function $v(t, z)$ is solved via backward time stepping. For the convection term in the operator $\mathcal{L}^a v$, an upwind difference scheme is adopted to ensure the monotonicity of the scheme (M-matrix property), a critical condition for convergence in stochastic control problems (Kushner, 1990):

$$\partial_z v(z_i) \approx \begin{cases} \frac{v_{i+1} - v_i}{\Delta z} & \text{if } \mu + a(t_n, z_i) \geq 0 \\ \frac{v_i - v_{i-1}}{\Delta z} & \text{if } \mu + a(t_n, z_i) < 0 \end{cases}$$

The optimal stopping constraint is enforced via a projection operation at each time step, similar to the Projected Successive Over-Relaxation (PSOR) method for obstacle problems (Glowinski, 2013):

$$v_i^n = \max \{ \tilde{v}_i^n, K(t_n, u^n) - C \}$$

where $K(t_n, u^n)$ is the endogenous outside option computed using the current matching control $u(t_n)$. The optimal stopping boundary $b^*(t_n)$ is identified as the critical state separating the continuation and stopping regions.

(3) Forward Solution of the Non-Local FPE Unlike standard decoupled frameworks, our population dynamics involve a *feedback loop* via the boundary flux. Once $b^*(t)$ is obtained, we solve the Non-Local Fokker–Planck Equation forward in time using a conservative *Fully Implicit Scheme*.

Flux Calculation via Mass Truncation: Instead of computing the gradient at the boundary, we calculate the swipe flux Φ_{swipe} by explicitly integrating the probability mass that falls

into the stopping region $z < b^*(t_n)$ during the implicit step. This approach aligns with the conservative flux reconstruction methods proposed by [Chang and Cooper \(1970\)](#):

$$\Phi_{\text{swipe}}(t_n) = \frac{1}{\Delta t} \sum_{z_i < b^*(t_n)} f^*(t_n, z_i) \Delta z$$

This mass is physically removed from the system and a fraction $(1 - \epsilon)$ is recycled back via the source term $g(z; u)$. This ensures rigorous mass conservation and stability.

(4) Iterative Resolution of the QVI Fixed Point Since the obstacle $K(t, u)$ depends on the value function v itself, a fixed-point iteration is required ([Bensoussan & Lions, 1984](#)):

- **Initialization:** Guess an initial trajectory for $K^{(0)}(t)$.
- **Update Step:** In iteration m , solve HJB backward to get $v^{(m)}$, then update the coupling variable:

$$K_{\text{new}}(t) = \int_{z_{\min}}^{z_{\max}} v^{(m)}(t, z) g(z; u(t)) dz$$

- **Convergence:** Apply relaxation until $\|K^{(m+1)} - K^{(m)}\|_{\infty} < \text{Tol}$.

B.2 Numerical Robustness and Convergence

To ensure that our equilibrium results are not artifacts of the numerical discretization or the domain truncation, we performed rigorous sensitivity analyses, as illustrated in [Figure 11](#).

Domain Independence (Panel A). We first verify the validity of the reflecting boundary condition at $z_{\max}$. We extended the computational domain from the baseline $z_{\max} = 10$ to $z_{\max} = 20$ while strictly maintaining the grid resolution Δz . As shown in [Figure 11\(a\)](#), the resulting optimal threshold curves $b^*(t)$ overlap closely. This confirms that the choice of $z_{\max} = 10$ is sufficiently large to eliminate boundary reflection artifacts.

Grid Convergence (Panel B). We explicitly test for discretization errors by refining the spatial grid $N_z \in \{400, 800, 1600\}$ while fixing $z_{\max} = 10$. [Figure 11\(b\)](#) demonstrates convergence of the solution. Notably, the solutions for $N_z = 800$ and $N_z = 1600$ are virtually indistinguishable, indicating that the solver reaches mesh independence at $N_z \approx 800$. While the baseline resolution used in exploratory simulations ($N_z = 400$) exhibits a minor quantitative offset due to discretization viscosity, it preserves the qualitative structure and dynamic trends of the converged solution.

B.3 Parameter Configuration and Pseudocode

The model parameters are calibrated to ensure a stable S-curve growth trajectory and a meaningful stationary equilibrium. [Table 5](#) details the specific settings used in the simulation. [Algorithm 1](#) presents the detailed numerical procedure.

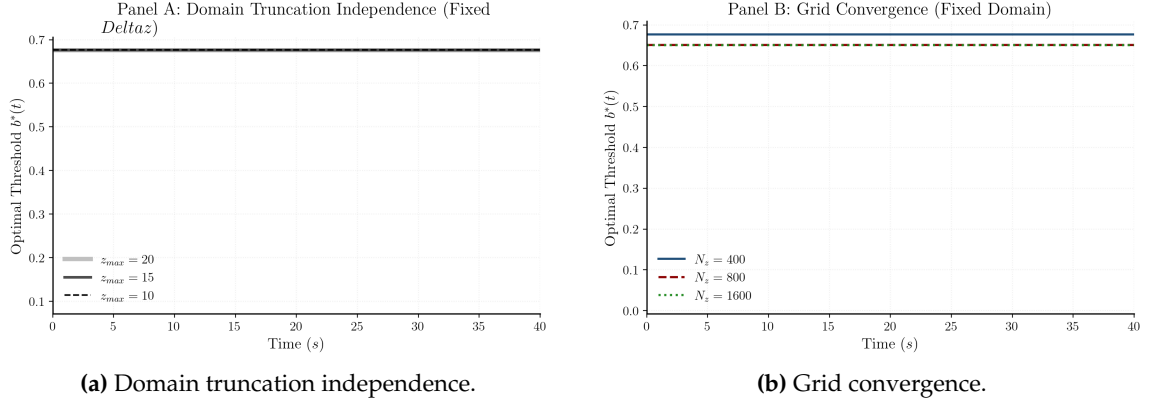


Figure 11: **Numerical robustness and convergence checks.** Panel (a) holds Δz fixed while expanding the computational domain ($z_{\max} = 10, 15, 20$) and shows that $b^*(t)$ is insensitive to domain truncation. Panel (b) fixes $z_{\max} = 10$ and refines the grid ($N_z = 400, 800, 1600$), demonstrating mesh convergence of the threshold dynamics.

Table 5: Model Parameters (Calibrated for Stability)

Symbol	Value	Description
<i>User & Content Parameters</i>		
γ	2.1	Relative risk aversion
ρ	0.5	Time discount rate (High impatience)
μ	-0.0235	Natural attention decay (Boredom)
σ	0.3	Content volatility
C	1.0	Cognitive switching cost
<i>Ecosystem Parameters</i>		
ϵ	0.0125	Churn Rate (1.25% exit probability)
A	5.0	Base Acquisition Rate
$N_{\max}$	100.0	Theoretical Market Capacity
μ_g	1.65	Content Quality Mean
<i>Numerical Settings</i>		
T	50.0	Simulation Time Horizon
Δt	0.01	Time Step Size
N_z	400	Uniform grid on $[0, 10]$ ($\Delta z \approx 0.025$)

Note: Parameters are calibrated to ensure a stable S-curve growth trajectory.

Algorithm 1 Fully Implicit Solver for User-Population Equilibrium

```
1: procedure SOLVEEQUILIBRIUM(Policy  $\{a, u\}$ , Params  $A, \epsilon, N_{\max}$ )
2:   Input: Control  $\{a, u\}$ , Market Params. Output:  $v^*, b^*, f^*, \Phi$ 
3:   Phase 1: User Optimization (Backward HJB)
4:   Construct implicit operators; Initialize implicit system.
5:   for  $n \leftarrow N_t - 1$  downto 0 do
6:     Solve linear system for  $\tilde{v}$ ;  $v^n \leftarrow \max(\tilde{v}, K(t_n) - C)$ 
7:      $b^*(t_n) \leftarrow \min\{z \mid v(t_n, z) > K(t_n) - C\}$  ▷ Identify threshold
8:     if Endogenous  $K$  then Update  $K(t_n)$  via fixed-point iteration
9:     end if
10:  end for
11:  Phase 2: Population Dynamics (Forward FPE)
12:  Initialize  $f(0, z)$  and  $N(0)$ ; Construct Fully Implicit Matrix  $\mathbf{M}_{FPE}$ 
13:  for  $n \leftarrow 0$  to  $N_t - 1$  do
14:    1. Acquisition:  $\alpha \leftarrow \max(0, 1 - N^n/N_{\max})$ ;  $S_{new} \leftarrow \alpha A \delta(z - z_p)$ 
15:    2. Implicit Solve:  $\mathbf{M}_{FPE} \cdot \mathbf{f}^* = \mathbf{f}^n + \Delta t \cdot S_{new}$ 
16:    3. Flux & Cut:  $M_{stop} \leftarrow \sum_{z < b^*} \mathbf{f}^* \Delta z$ ;  $\Phi(t_n) \leftarrow M_{stop}/\Delta t$ 
17:    Set  $\mathbf{f}^*(z) = 0$  for all  $z < b^*(t_{n+1})$  ▷ Truncate distribution
18:    4. Recycling:  $\mathbf{f}^{n+1} \leftarrow \mathbf{f}^* + (1 - \epsilon)\Phi(t_n)g(z)\Delta t$ ; Update  $N(t_{n+1})$ 
19:  end for
20:  return  $\{v, b, f, \Phi\}$ 
21: end procedure
```

C Derivation of the Master Equation on the Space of Measures

In this appendix, we derive the Hamilton-Jacobi-Bellman (HJB) equation for the platform's value function $V(N, m)$, utilizing the structure-scale decomposition introduced in Section 3.3.1.

C.1 Variational Derivatives and Chain Rule

Let $\mathcal{W}(f)$ be the value functional on the space of unnormalized measures. We decompose $f = N \cdot m$. To establish the Master Equation in terms of (N, m) , we first relate the functional derivative $\frac{\delta \mathcal{W}}{\delta f}$ to the derivatives with respect to N and m .

Consider a perturbation $f \rightarrow f + \delta f$. The variations in N and m are:

$$\delta N = \int \delta f(z) dz, \quad \delta m(z) = \frac{\delta f(z)}{N} - \frac{f(z)}{N^2} \delta N.$$

The total variation of the value function is:

$$\begin{aligned} \delta \mathcal{W} &= \frac{\partial V}{\partial N} \delta N + \int \frac{\delta V}{\delta m}(z) \delta m(z) dz \\ &= \int \left[\frac{\partial V}{\partial N} + \frac{1}{N} \left(\frac{\delta V}{\delta m}(z) \right. \right. \\ &\quad \left. \left. - \int \frac{\delta V}{\delta m}(y) m(y) dy \right) \right] \delta f(z) dz. \end{aligned} \tag{46}$$

Thus, the shadow price of the unnormalized density is given by the *Decomposition Identity*:

$$\begin{aligned}\frac{\delta \mathcal{W}}{\delta f}(z) &= \frac{\partial V}{\partial N} + \frac{1}{N} \hat{\lambda}(z), \\ \text{where } \hat{\lambda}(z) &= \frac{\delta V}{\delta m}(z) - \mathbb{E}_m \left[\frac{\delta V}{\delta m} \right].\end{aligned}\tag{47}$$

C.2 The Bellman Principle

The dynamic programming principle states:

$$\begin{aligned}rV(N, m) &= \sup_{a, u, A} \left\{ \pi(N, m, a, u, A) \right. \\ &\quad \left. + \frac{\partial V}{\partial N} \frac{dN}{dt} + \left\langle \frac{\delta V}{\delta m}, \frac{\partial m}{\partial t} \right\rangle \right\}\end{aligned}\tag{48}$$

Substituting the dynamics derived in the main text:

- **Scale Dynamics:** $\frac{dN}{dt} = A - \epsilon N \phi_{swipe}(m)$
- **Structural Dynamics:** $\frac{\partial m}{\partial t} = \mathcal{L}^*[a]m + \frac{A}{N}(\delta_{z_p} - m) + (1 - \epsilon)\phi_{swipe}(m)g + \epsilon \phi_{swipe}(m)m$

C.3 Optimization of Controls

Before optimizing specific controls, we must explicitly handle the duality pairing involving the differential operator $\mathcal{L}^*[a]$.

Green's Identity and Boundary Terms: The term $\langle \frac{\delta V}{\delta m}, \mathcal{L}^*[a]m \rangle$ involves the Fokker-Planck operator. Using Green's identity (integration by parts twice) on the domain $\Omega = [b^*, z_{max}]$ with the absorbing boundary condition $m(b^*) = 0$:

$$\begin{aligned}\int_{b^*}^{z_{max}} \frac{\delta V}{\delta m} \mathcal{L}^*[a]m \, dz &= \int_{b^*}^{z_{max}} \mathcal{L}^a \left[\frac{\delta V}{\delta m} \right] m \, dz \\ &\quad - \underbrace{\frac{\delta V}{\delta m}(b^*) \cdot \frac{1}{2} \sigma^2(b^*)^2 \partial_z m(b^*)}_{\phi_{swipe}(m)}\end{aligned}\tag{49}$$

The boundary term $-\frac{\delta V}{\delta m}(b^*)\phi_{swipe}(m)$ arises because while the density m vanishes at the absorbing boundary, its gradient (flux) is non-zero. This term represents the *value destruction* of a user hitting the churn threshold. Note that we assume zero flux at the reflective boundary z_{max} .

We now maximize the Hamiltonian term by term:

1. Retention Control $a(z)$: The drift control $a(z)$ appears in the generator $\mathcal{L}^a[\frac{\delta V}{\delta m}] = (\mu + a(z))z\partial_z(\frac{\delta V}{\delta m}) + \dots$. Combining with the cost term $-\frac{\xi_1}{2}N \int a^2 m dz$:

$$\sup_a \int \left(-\frac{\xi_1}{2} N a^2 + a z \partial_z \frac{\delta V}{\delta m} \right) m dz \implies a^*(z) = \frac{z}{\xi_1 N} \partial_z \frac{\delta V}{\delta m}$$

2. Acquisition Control A : Terms involving A :

$$A \left(\frac{\partial V}{\partial N} + \frac{1}{N} \frac{\delta V}{\delta m}(z_p) - \frac{1}{N} \int \frac{\delta V}{\delta m} m dz \right) - \Psi(A)$$

Using the identity (47), the coefficient of A is simply $\frac{\delta \mathcal{W}}{\delta f}(z_p)$. The maximum is $H_{acq}(\frac{\delta \mathcal{W}}{\delta f}(z_p))$.

3. Matching Control u : Terms involving u (via $g(z; u)$ and costs) and the flux ϕ_{swipe} (which collects the revenue, recycling, scale loss, and boundary value loss terms):

$$\begin{aligned} \text{Terms} = \phi_{swipe} & \left[\underbrace{\lambda_2 N}_{\text{Ads}} + \underbrace{(1 - \epsilon) \int \frac{\delta V}{\delta m} g(z; u) dz}_{\text{Recycling Gain}} \right. \\ & - \underbrace{(1 - \epsilon) \int \frac{\delta V}{\delta m} m dz}_{\text{Mean Value Correction}} - \underbrace{\epsilon N \frac{\partial V}{\partial N}}_{\text{Churn Loss}} \\ & \left. - \underbrace{\frac{\delta V}{\delta m}(b^*)}_{\text{Boundary Loss}} \right] - \frac{\xi_2}{2} u^2 \end{aligned} \quad (50)$$

Let $\Delta V_{swipe}(u)$ denote the net marginal value of a swipe (the term in brackets). The optimal matching u^* satisfies:

$$\xi_2 u^* = \phi_{swipe} (1 - \epsilon) \int \frac{\delta V}{\delta m} \frac{\partial g}{\partial u}(z; u^*) dz$$

C.4 The Final Equation

Collecting all terms yields the explicit Master Equation presented in Eq. (30). The inclusion of the boundary term $-\frac{\delta V}{\delta m}(b^*)\phi_{swipe}$ ensures the mathematical consistency of the Mean-Field Stackelberg Game on a domain with absorbing boundaries.

D Numerical Solution Details

D.1 MDP Formulation and Spectral Action Embedding

We reformulated the Mean-Field Stackelberg Game (MFSG) as a Markov Decision Process (MDP) defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ as follows:

State Space $\mathcal{S}$. The state at time t is represented by the macroscopic configuration of the system, comprising the total population and the discretized density distribution of user attention:

$$\mathbf{s}_t = [N(t), m(t, z_1), m(t, z_2), \dots, m(t, z_{N_z})]^\top \in \mathbb{R}^{N_z+1} \quad (51)$$

where $\{z_i\}_{i=1}^{N_z}$ are the spatial grid points. This high-dimensional state input allows the neural network to capture the fine-grained structure of the population distribution (Carmona & Laurière, 2021).

Spectral Action Space $\mathcal{A}$. A core challenge is that the action $a(t, \cdot)$ is a function in an infinite-dimensional space $L^2(\Omega)$. To mitigate the *curse of dimensionality* (Bellman, 1966), we employ a spectral parameterization method. We project the control function onto a subspace spanned by the first K Chebyshev polynomials $T_k(\cdot)$ (Boyd, 2001):

$$a(t, z) \approx \sum_{k=0}^{K-1} w_k(t) T_k(\tilde{z}), \quad \tilde{z} = \frac{2z - z_{max}}{z_{max}} \in [-1, 1] \quad (52)$$

Consequently, the action vector output by the agent is finite-dimensional:

$$\mathbf{u}_t = [u(t), A(t), w_0(t), \dots, w_{K-1}(t), u_{match}(t), \sigma_g(t)]^\top \in \mathbb{R}^{K+4} \quad (53)$$

where $u_{match}(t)$ controls content matching quality and $\sigma_g(t)$ controls content diversity. This spectral embedding reduces the action dimension from $N_z \approx 400$ to $K + 4 = 14$ (with $K = 10$), while naturally enforcing the spatial smoothness of the algorithmic intervention $a(t, z)$, which is crucial for the stability of the HJB solver.

Reward Function $\mathcal{R}$. The immediate reward reflects the platform’s instantaneous profit flow:

$$r_t = N(t) (\lambda_1 \bar{z}_{ret} + \lambda_2 \Phi_{swipe}) - \left(\frac{\xi_1}{2} \|a\|_2^2 + \frac{\xi_2}{2} u^2 + \Psi(A) \right) \quad (54)$$

where $\bar{z}_{ret}$ is the average retention utility and Φ_{swipe} is the flux of swiping users.

D.2 Algorithm: Spectral Soft Actor-Critic

We adopt the Soft Actor-Critic (SAC) algorithm (Haarnoja, Zhou, Abbeel, & Levine, 2018), an off-policy maximum entropy DRL method. The interaction between the Agent (Platform) and the Environment (User Population) is governed by a physics-based engine that solves the coupled HJB-FPK system at each time step. The complete solution procedure is detailed in Algorithm 2.

D.3 Computational Infrastructure

To ensure the reproducibility of our numerical results, we detail the hardware and software environment used for the simulations and reinforcement learning training processes.

Hardware Specifications. All computations were executed on a high-performance computing server equipped with:

- **GPU:** NVIDIA Tesla P100 (16GB VRAM), ensuring efficient parallel computation for the neural network updates.
- **CPU:** Intel Xeon Platinum 8260 (2.30GHz, 12 vCPUs).
- **Memory:** 60GB RAM to handle the storage of the Experience Replay Buffer (size 10^6 transitions).

Algorithm 2 Spectral SAC with Physics-Informed Environment

```
1: Input: Basis  $\{T_k\}_{k=0}^{K-1}$ ; Params  $N_{\max}, \epsilon$ ; Rates  $\alpha, \beta$  Output: Policy  $\pi_\phi^*$ , Value  $Q_\theta^*$ 
2: Initialize Actor  $\pi_\phi$ , Critic  $Q_\theta$ , and Replay Buffer  $\mathcal{D}$ 
3: for episode  $m = 1$  to  $M$  do
4:   Initialize state  $\mathbf{s}_0 = [N_0, \mathbf{f}_0]$ 
5:   for step  $t = 0$  to  $T_{\text{horizon}}$  do

6:     Phase 1: Spectral Action (Leader). Sample  $\mathbf{u}_t \sim \pi_\phi(\cdot | \mathbf{s}_t)$  and reconstruct field:
7:      $a(z) \leftarrow \sum_{k=0}^{K-1} w_k T_k(\tilde{z})$  where  $\mathbf{u}_t = [u, A, \{w_k\}]$  and  $\tilde{z} \in [-1, 1]$ 
8:     Phase 2: User Equilibrium (HJB). Solve QVI via fixed-point iteration:
9:     Find  $v(z)$  s.t.  $\min(\mathcal{L}v - \rho v + U, v - (K - C)) = 0$ 
10:    Identify threshold:  $b_t^* \leftarrow \min\{z \mid v(z) > K(u) - C\}$ 

11:    Phase 3: Population Dynamics (Implicit FPE).
12:    1. Acquisition:  $S_{\text{new}} \leftarrow \max(0, 1 - N_t/N_{\max}) \cdot A \cdot \mathcal{N}(z)$ 
13:    2. Implicit Solve:  $\mathbf{M}_{FPE} \cdot \mathbf{f}^* = \mathbf{f}_t + \Delta t \cdot S_{\text{new}}$ 
14:    3. Flux & Cut:  $\Phi_t \leftarrow (\sum_{z < b_t^*} \mathbf{f}^* \Delta z) / \Delta t$ ; Set  $\mathbf{f}^*(z) \leftarrow 0$  for  $z < b_t^*$ 
15:    4. Recycling:  $\mathbf{f}_{t+1} \leftarrow \mathbf{f}^* + (1 - \epsilon)\Phi_t g(z; u)\Delta t$ ; Update  $N_{t+1}$ 

16:    Phase 4: Learning. Store  $(\mathbf{s}_t, \mathbf{u}_t, r_t, \mathbf{s}_{t+1})$  in  $\mathcal{D}$ .
17:    Update Critic  $\theta$  (Bellman) and Actor  $\phi$  (Max Entropy  $J = \mathbb{E}[Q - \alpha \log \pi]$ ).
18:  end for
19: end for
20: return Optimized Policy  $\pi_\phi^*$ 
```

Software Stack. The solution algorithm was implemented using **Python 3.10** and **PyTorch 2.0.1** with CUDA 11.8 support. The Mean-Field Stackelberg Game solver and the Soft Actor-Critic (SAC) agent were built upon the *Gymnasium* interface standards and the *Stable Baselines 3* library.

D.4 Hyperparameter Configuration and Reward Shaping

To ensure the stability of the Spectral SAC algorithm and accurate calibration against the empirical moments, we employed specific reward shaping coefficients. The training process utilized the Stable Baselines 3 implementation of Soft Actor-Critic. Table 6 details the key hyperparameters for both experiments. Both configurations were trained for 750,000 steps per seed across 10 independent random seeds.

Remark 6 (Reward Shaping as Exploration Anchor, Not Hypothesis Generator). A potential concern is that the flux-floor, penalty, and bonus terms in Table 6 may themselves *generate* the high-flux Clickbait result rather than merely help the optimizer *discover* it. We address this concern on four grounds.

First, both experiments employ flux shaping, not just Clickbait. The Baseline uses flux floor = 0.5 and penalty = 1.0 (bonus = 0.0); the Clickbait uses floor = 1.5, penalty = 1.5, bonus = 2.0. The design is symmetric in kind; Clickbait values are larger because the optimization landscape near the $\mathcal{R} = 1$ bifurcation is far harder to explore with a 14-dimensional Chebyshev action space. Without a minimum flux signal, SAC receives zero gradient on the flux dimension during early training and fails to explore flux-generating

Table 6: Reward Function Coefficients and Hyperparameters

Parameter	Baseline	Clickbait	Description
<i>Experiment-Specific Parameters (Matching Eq. 25)</i>			
λ_1	20.0	5.0	Weight on retention utility flow
λ_2	0.1	5.0	Weight on swipe flux
Flux floor	0.0	1.5	Minimum flux reward threshold
Flux floor penalty	0.0	1.5	Penalty for falling below flux floor
Flux bonus	0.0	2.0	Bonus for exceeding flux floor
c_{ctrl}	2.0	3.0	Control cost scaling coefficient
c_{smooth}	5.0	6.0	Spectral smoothness penalty
<i>Shared Structural Parameters</i>			
ξ_1	1.0		Retention control cost ($\frac{\xi_1}{2} \ a\ ^2$)
ξ_2	0.5		Matching investment cost ($\frac{\xi_2}{2} u^2$)
ψ_0	0.01		Acquisition cost coefficient
<i>SAC Training Configuration (Both Experiments)</i>			
Total Steps	750,000		Per seed
Seeds	10		Independent runs
Learning Rate	3×10^{-5}		Actor & Critic (Adam)
Batch Size	512		Experience replay
γ	0.99		Discount factor

Note: λ_1, λ_2 correspond to the objective function in Eq. (25). In the implementation, these are scaled by a common multiplier for numerical stability (the ratio $\mathcal{R} = \lambda_2/\lambda_1$ is preserved). Both experiments employ flux-floor shaping; the Clickbait values are larger because the steeper optimization landscape near the $\mathcal{R} = 1$ bifurcation requires stronger exploration incentives. See Remark 6 for a full discussion of the role of these terms. c_{ctrl} and c_{smooth} are numerical regularization coefficients that prevent degenerate policies; they do not appear in the theoretical model.

policies at all—this is an exploration problem, not a hypothesis.

Second, the qualitative contrast cannot be explained by shaping magnitude differences. The Baseline and Clickbait outcomes are separated by two orders of magnitude in flux and nearly 60% in population. A shaping coefficient difference of $\sim 2\times$ cannot plausibly produce this gap; the gap is consistent only with the structural difference in $\mathcal{R}$.

Third, and most critically, the PDE equilibrium scan (Section 6.3) is entirely free of reward shaping. That scan integrates the non-local Fokker-Planck equation directly across a grid of b^* values with $a(z) \equiv 0$ and no SAC optimizer. It produces the same qualitative finding—dwell time $\tau^*(b^*)$ crosses the IAB/MRC threshold at $b_{\text{crit}}^* \approx 1.7$ —using no reward shaping parameters whatsoever. The DRL results are corroborating evidence; the PDE scan is the primary, model-driven evidence for the trap.

Fourth, the scale-invariance robustness check (Appendix E) trains a matched-budget experiment (Experiment C: $\lambda_1 = \lambda_2 = 10$) and obtains the same high-flux equilibrium, confirming that the result is determined by the ratio $\mathcal{R} = 1$, not by the absolute magnitude of any reward component.

E Robustness Check: Scale Invariance and Algorithmic Overheating

To rigorously verify that the “Algorithmic Liquidity Trap” identified in Experiment A is not a numerical artifact of lower absolute reward weights, we performed a normalization experiment (Experiment C).

Experimental Design. We constructed a scenario that matches the *Total Weight Budget* of the Baseline (≈ 20) but retains the *Monetization Ratio* of the Ad-Centric regime ($\mathcal{R} = 1$).

- **Baseline:** $\lambda_1 = 20.0, \lambda_2 = 0.1$ (Total ≈ 20).
- **Experiment A (Original):** $\lambda_1 = 5.0, \lambda_2 = 5.0$ (Total = 10).
- **Experiment C (Robustness):** $\lambda_1 = 10.0, \lambda_2 = 10.0$ (Total = 20).

Results and Interpretation. The results are visualized in Figure 12.

1. **Validation of the Trap (Flux Consistency):** The equilibrium swipe flux Φ^* in Experiment C converged to a value consistent with Experiment A, representing a multiple-fold increase over the Baseline. This confirms that the high-velocity trap is a structural equilibrium determined by the ratio $\mathcal{R}$, robust to parameter scaling.
2. **Algorithmic Overheating (Population Drop):** The steady-state population N^* in Experiment C was lower than in Experiment A, a structurally consistent outcome of *Relative Cost Reduction*. In the objective function $J = \lambda_1 z + \lambda_2 \Phi - \frac{\xi}{2} a^2$, doubling the rewards (λ) while holding the control cost coefficient constant ($\xi = 1.0$) effectively makes algorithmic intervention “cheaper” relative to the potential payoff. Consequently, the DRL agent adopted a more aggressive “harvesting” policy—pushing the

recycling velocity to its limit to extract immediate flux rewards, even at the cost of higher user burnout and a lower carrying capacity. We term this phenomenon “*Algorithmic Overheating*,” where excessive incentive intensity (without corresponding cost constraints) accelerates ecosystem depletion. This aggressive harvesting drives the implicit dwell time τ down even further below the IAB/MRC viewability threshold, indicating that higher reward scales without cost adjustments exacerbate the race to the bottom.

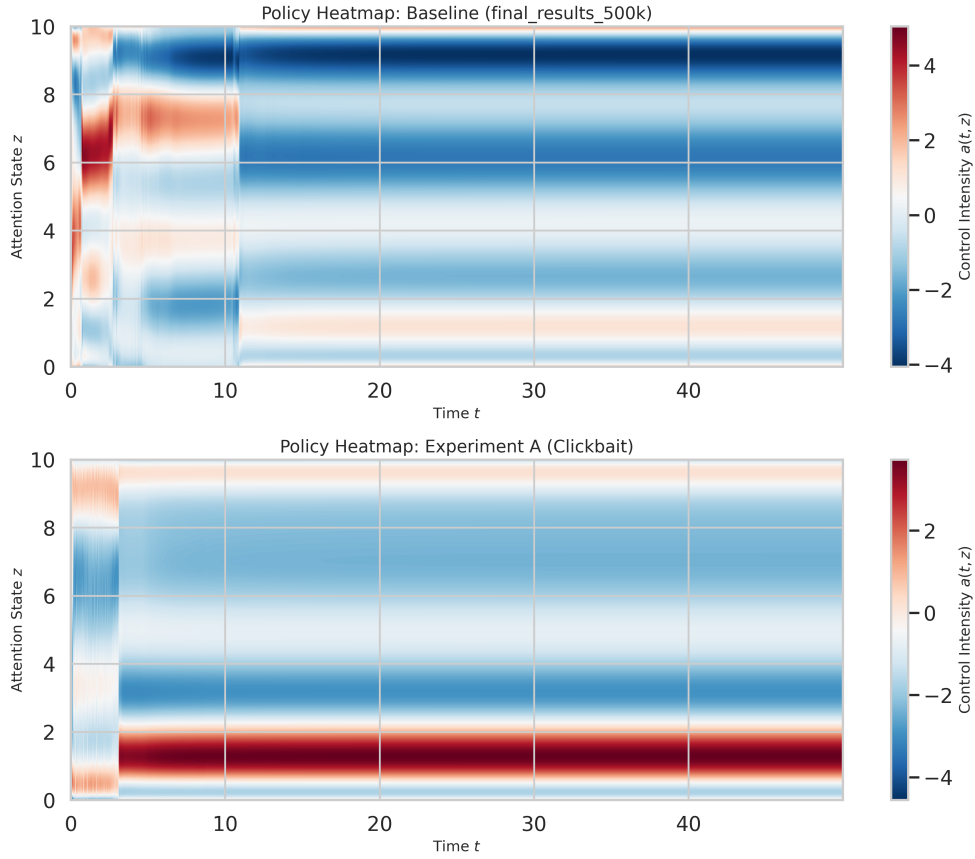


Figure 12: **Robustness Check: Effect of Reward Scale Normalization.** Comparing Experiment A ($\lambda = 5, 5$) with the scaled Experiment C ($\lambda = 10, 10$). **Right Bar:** The Swipe Flux (Φ^*) remains robustly high (> 100), confirming the persistence of the Liquidity Trap regardless of scale. **Left Bar:** The drop in Population (N^*) in Exp C illustrates the “Overheating” effect, where relatively cheaper control costs lead to aggressive over-harvesting.